

One Markov Chain Monte Carlo Fit for Many Plausible Values: A Calibrated Stacked Posterior Workflow for Bayesian Multilevel Models of Large-Scale Assessment Data

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ABSTRACT. Large-scale assessments such as the Programme for International Student Assessment (PISA) report each student’s proficiency as multiple plausible values (PVs), latent traits published as draws from a marginal posterior distribution. The PVs, in turn, are analyzed by data users through fitting a secondary model once per PV and pooling the results with the combining rules of multiple imputation. For a Bayesian multilevel model, this convention requires one Markov chain Monte Carlo (MCMC) run per plausible value. We develop a workflow in which a single MCMC run suffices. All plausible values enter one stacked fit, and the reported estimate and total variance are assembled from deterministic per-PV fits. This paper formulates an exact affine correction and calibrates the fixed-effect posterior draws to the assembled target. A simulation across multiple conditions shows accurate estimation and near-nominal interval coverage. In an application to PISA 2022 reading data for the United States and Korea, the recommended workflow reproduces the per-PV estimates and standard errors. The open-source R package `pystackr` implements the workflow.

Keywords: plausible values; large-scale assessment; Bayesian multilevel model; multiple imputation; PISA

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1. Introduction

Large-scale assessments such as the Programme for International Student Assessment (PISA) administer to each student only a small sample of items from the assessment’s item pool. Individual proficiencies, therefore, cannot be estimated with useful precision. The assessment characterizes each student’s proficiency as a latent trait estimated from an item-response model. The model outputs a conditional distribution for each student’s latent trait, θ_i , given the observed item responses, $\mathbf{y}_i$, and background variables, $\mathbf{x}_i$, after integrating over, or fixing at estimates of, the parameters of the item-response model. Rather than collapse that distribution into a single score, which would discard the measurement uncertainty, the assessment program draws from it repeatedly and releases the draws, indexed $m = 1, \dots, M$. These draws are labeled as *plausible values* (PVs) and are published to data users in lieu of the student test scores: in PISA, ten alternative draws of every student’s proficiency are published. Each proficiency draw, along with available predictors, is labeled a “completion” (Mislevy, 1991; Mislevy et al., 1992; OECD, 2024b; Wu, 2005). The data files of latent traits carry a standing instruction for secondary analysis: the analysis model is to be fit once to each plausible value, and the results are to be pooled with the combining rules of multiple imputation (Jewsbury et al., 2024; Laukaityte & Wiberg, 2017; Rubin, 1987; Rutkowski et al., 2010). The pooled point estimate averages the per-PV estimates, and the pooled variance adds to the average within-PV variance a between-PV component that reflects how much the estimate varies across realizations of the measurement uncertainty. For the commonly used least-squares regressions, fitting these multiple models for each PV is computationally fast and simple.

Data users typically perform inference on the published PVs through a secondary model, referred to throughout as the *analysis model*. The analysis models of contemporary school-effects research impose a higher computational cost because they are more richly parameterized. A researcher studying socioeconomic gradients in PISA now commonly fits a Bayesian multilevel model, with students nested in schools, survey weights entering the likelihood, and estimation by Markov chain Monte Carlo (MCMC). For such a model, the combining convention requires M separate sampler runs, M sets of convergence diagnostics, and M fitted model objects, and this entire computational cost recurs with every model revision, sensitivity analysis, and additional country in the study. The convention of fitting M separate models remains statistically sound, but its computational burden falls most heavily on the iterative model development that Bayesian practice encourages (Kaplan & Harra, 2024). This

article develops a procedure that allows a single MCMC run to deliver the same resulting point and variance estimates that the M -run convention delivers.

The workflow we develop is motivated by the observation that a full Bayesian model fit, with its posterior sample, is not required for each PV draw in order to obtain a pooled estimate under the multiple-imputation combining rules. The pooled point estimate and total variance over the completions can be computed without a full posterior distribution. We use a secondary statistical model to obtain a joint posterior distribution for the model’s parameters, but implementing the combining rules requires only per-PV statistics, and the per-PV results can be produced by deterministic fits that involve no MCMC sampling. The proposed workflow therefore assigns each task to the tool suited to it:

- (1) A single *stacked* analysis model is estimated by combining all M plausible values at once, each PV’s likelihood entering with fractional weight $1/M$, and leading to a single posterior distribution.
- (2) The target point and variance are assembled using the combining rules with *deterministic* per-PV fit statistics.
- (3) An affine adjustment is applied once to the fixed-effect draws from the posterior distribution from the *stacked* analysis model. This calibrates the mean and covariance of the analysis model to equal the target values for the mean and variance.

We refer to the resulting multi-step procedure as the *calibrated-stack workflow*. We present a theoretical result in Section 4 that shows that, under a common working covariance, the fixed-effect estimate from the stacked approach equals the pooled per-PV estimate rather than approximating it.

We conduct a simulation study with 24 combinations of conditions that are commonly encountered in analyses of large-scale assessments like PISA: intraclass correlations, school counts, school sizes, PV reliabilities, and PV counts. Results show that the calibrated stack recovers fixed-effect estimates and covers at close to the nominal rate, matching the per-PV workflow benchmark. The PISA 2022 application to the United States and Korea examines the calibrated stack and the reweighting alternative under one analysis protocol. The calibrated stack reproduces the per-PV workflow’s estimates and standard errors in both countries. A reweighting alternative performs well in the United States but not in Korea. This contrast motivates our recommendation to calibrate rather than reweight.

The article makes three contributions. First, it states and proves the fixed-effect point identity under the more computationally scalable stacked fit procedure, which demonstrates that our single model run procedure achieves the target point estimate. Second, this paper develops a calibration step for the stacked fit as an exact, single-step transfer of an externally assembled target onto the stacked draws. Third, this paper evaluates the multistep workflow in a designed simulation and demonstrates it end-to-end on PISA 2022 reading data for the United States and Korea. The calibrated stack reproduces the per-PV workflow’s estimates and standard errors for the within- and between-school socioeconomic-status effects. The open-source R package `pystackr` implements every step (Section 8).

The remainder of the article is organized as follows. Section 2 reviews plausible values and the combining convention that motivates the design of our proposed workflow. Section 3 describes Bayesian multilevel modeling for assessment data and related work on ‘one-fit’ inference. Section 4 develops the three alternative workflows. Sections 5 and 6 assess the workflows through a simulation study and the PISA application. Section 7 concludes with a discussion of practical guidance and limitations.

2. Plausible Values and the Combining Convention

Plausible values are multiple imputations of a latent trait under a primary model. Because a short test cannot determine a student’s proficiency precisely, the assessment treats proficiency as missing data: an item response model and a latent regression on background variables define each student’s conditional proficiency distribution, $f(\theta_i | \mathbf{y}_i, \mathbf{x}_i)$, after integrating over the parameters of the item-response model; operational scaling holds the item parameters at their estimates and draws the latent-regression coefficients from their estimated distribution (OECD, 2024b). M draws from this posterior distribution are released in lieu of the student test scores (Mislevy, 1991; Rubin, 1987). Each posterior draw is an unbiased realization with respect to the population model, but no single draw can carry the measurement uncertainty itself, which is expressed in the variation across draws. The combining rules restore that variation to the reported standard errors.

Concretely, the analyst fits the same analysis model to the latent-trait draws M times, once per PV, obtaining fixed-effect estimates $\hat{\beta}_m$ and within-fit covariance estimates $\hat{U}_m$, and reports

$$\bar{\beta}_{\text{MI}} = \frac{1}{M} \sum_{m=1}^M \hat{\beta}_m, \quad T_{\text{MI}} = \bar{U} + \left(1 + \frac{1}{M}\right)B, \quad (1)$$

where $\bar{U} = M^{-1} \sum_m \hat{U}_m$ is the average within-imputation covariance and $B = (M - 1)^{-1} \sum_m (\hat{\beta}_m - \bar{\beta}_{\text{MI}})(\hat{\beta}_m - \bar{\beta}_{\text{MI}})^\top$ is the between-imputation covariance (Rubin, 1987, 1996). Interval estimates use a student- t reference with the Barnard and Rubin (1999) degrees of freedom. The convention presumes a congenial pairing of primary and secondary analysis models (Meng, 1994; Murray, 2018), and it has supported several decades of applied practice.

One structural property of Equation (1) motivates the design of the workflow proposed in this article. The two variance components have different sources of variation. The within component $\bar{U}$ averages the M covariance estimates $\hat{U}_m$, each of which measures the sampling uncertainty of one analysis given one completed dataset (that includes a draw from the posterior distribution of θ_i in the primary model). The between component B is a covariance of the M estimates $\hat{\beta}_1, \dots, \hat{\beta}_M$ estimated on the M completed datasets: B measures how the estimate moves when the completion of the data changes and everything else is held fixed. Information about B therefore resides only in contrasts between analyses of different completions, and no functional of a single analysis contains it. Appendix C.4 of the online supplement records the consequence for replication schemes applied to a single fit, which perturb the sampled units while holding the completions fixed and therefore recover only the within component $\bar{U}$. The quantities in Equation (1) are instead assembled from deterministic per-PV computations that require no MCMC, and the single Bayesian model estimated on the stacked data supplies the posterior draws that the workflow calibrates (Section 4).

3. Review of Bayesian Multilevel Secondary Analysis Models

The motivating model for this article is the class of two-level random-intercept regressions of achievement on student- and school-level covariates. These regressions are applied as a secondary analysis to plausible values produced by primary assessment models. Such analyses are widely used in educational assessment research (e.g. Rabe-Hesketh & Skrondal, 2022).

3.1. Accounting for the Sampling Design. Assessment samples are drawn through a two-stage design, schools first followed by students within schools. Both schools and students may be selected with unequal probabilities, which requires model-based analyses to incorporate the sampling weights in the likelihood (Pfeffermann et al., 1998; Rabe-Hesketh & Skrondal, 2006). We follow the survey-weighted pseudo-posterior approach (Savitsky & Toth, 2016; Savitsky & Williams, 2022;

Williams & Savitsky, 2020; Williams et al., 2025): student i ’s log-likelihood contribution is scaled by a sampling weight normalized to sum to the sample size within each country (equivalently, to mean one); this Hájek-style normalization keeps the posterior’s concentration on the order of the realized sample size. The weighting adjusts the estimating equations for unequal, and potentially informative, inclusion probabilities; it is a pseudo-posterior adjustment aimed at student-population estimands, not a generative model for the sampling design. We use the assessment’s final student weight, which is constructed as the product of a school-stage factor and a within-school factor and therefore carries both stages of selection (OECD, 2024b).

Where the weights should enter a multilevel likelihood, particularly for variance components and school-level parameters, is a subtle question with a literature of its own (Asparouhov, 2006; Cai, 2013; Lumley & Huang, 2023; Mang et al., 2021; Pfeiffermann et al., 1998; Pfeiffermann, 1993; Rabe-Hesketh & Skrondal, 2006). We use the normalized final student weight at the student level (level 1), so that comparisons across the alternative workflows are not confounded by weighting choices. This choice is well suited to the fixed effects that are this article’s reportable quantities; the variance components, for which placement matters most (Pfeiffermann et al., 1998; Rabe-Hesketh & Skrondal, 2006), are treated as nuisance parameters throughout (Section 4.1). We consider the survey’s two-stage weight decomposition (weights for schools and weights for students within schools) as part of an alternative approach (Section 4.5). We also include it in a sensitivity analysis in the online supplement (Appendix F.4).

3.2. Alternative Secondary Model Workflows. Kaplan and Harra (2024) propose a Bayesian workflow for international assessment data. They note that an open problem is how the assessments’ replicate-weight machinery should enter Bayesian estimation. Analogously, the convention for using plausible values raises a question of the same kind. The established convention is to apply Equation (1) directly, with M complete posterior samples run independently and pooled at the end. We formalize this below as the per-PV workflow, and use it as the benchmark for comparison.

Efforts to modify this convention have taken two directions. Within the multiple-imputation literature, Zhou and Reiter (2010) studied posterior simulation that mixes draws across the completed datasets. They found that reliable summaries require more completions than the customary five to ten. Si and Reiter (2011) compared this route with combining-rule inference and found that, for missing-data inference, intervals based on the combining rules generally have better repeated-sampling properties,

with posterior simulation retaining an advantage only at high fractions of missing information. Si and Reiter (2013) developed fully Bayesian imputation modeling for the categorical background variables of a large-scale assessment and found that a Dirichlet-process mixture model captures dependencies among eighty TIMSS background variables that standard log-linear and sequential-regression imputation can fail to represent.

More recently, Jedhoff et al. (2025) proposed running MCMC for only one or a few representative imputations and recovering the remaining per-imputation posteriors from those draws by importance reweighting. Pareto-smoothed importance sampling (PSIS; Vehtari et al., 2024) provides both stabilization and a tail-shape diagnostic $\hat{k}$: values above 0.7 mark the reweighting as unreliable. Reweighting requires no external target, but it makes the reportability of every estimate contingent on the diagnostic: when $\hat{k}$ crosses the threshold, no estimates can be reported. The reweighted variant examined in this article (Section 4.5) instead uses a single posterior over all M stacked imputations as the proposal, which is a demanding assignment: each reweighting requires a geometric mixture to serve as proposal for one of its own components. Our stacked likelihood object falls within the class of fractional and composite likelihoods, which are themselves well studied (Bissiri et al., 2016; Holmes & Walker, 2017; Kleijn & van der Vaart, 2012; Koers et al., 2026; Ribatet et al., 2012). When the resulting posterior’s covariance is not the covariance one wishes to report, the remedy is to calibrate it to an explicit target rather than to reinterpret it (Hoff & Wakefield, 2012; Little, 2011; Müller, 2013; Williams & Savitsky, 2021).

Our proposed workflow combines these lines of work rather than replacing any of them. From multiple imputation it takes Equation (1) as its target point and variance estimates (Meng, 1994; Murray, 2018). From the survey-weighted pseudo-posterior tradition it takes the weighting convention above and the calibration idea, specializing the scale-and-shape correction of Williams and Savitsky (2021) to the fixed-effect block of a multilevel model with a combining-rule covariance target. From the one-fit reweighting line of Jedhoff et al. (2025) it takes the aim of serving every per-imputation analysis from a single set of posterior draws; the stacked fractional proposal itself is this article’s construction (Section 4.3). The point and variance estimation target itself is assembled externally, which is consistent with design-based software (Judkins, 1990; OECD, 2024b; Zhang et al., 2024). The next section develops the three workflows in full.

4. Three Workflows for Plausible-Value Inference

4.1. **Secondary Model and Estimands.** Let $\theta_{ij}^{(m)}$ denote the m th plausible value for student i in school j , and let x_{ij} denote a student-level covariate; in the application, x_{ij} is the PISA index of economic, social, and cultural status (ESCS). Decomposing the covariate into within- and between-school parts (Allison, 2009; Rabe-Hesketh & Skrondal, 2022), the analysis model for each PV is the two-level random-intercept regression

$$\theta_{ij}^{(m)} = \beta_0 + \beta^W(x_{ij} - \bar{x}_{.j}) + \beta^B \bar{x}_{.j} + \zeta_j + \varepsilon_{ij}^{(m)}, \quad \zeta_j \sim N(0, \tau^2), \quad \varepsilon_{ij}^{(m)} \sim N(0, \sigma^2), \quad (2)$$

where $\bar{x}_{.j}$ is the school mean of the covariate. The estimands are the fixed effects $\boldsymbol{\beta} = (\beta_0, \beta^W, \beta^B)^\top$: the within-school gradient β^W and the between-school gradient β^B carry the scientific interest. The variance components (τ^2, σ^2) are estimated in every workflow but treated as nuisance parameters; the reportable quantities in this article are the fixed effects.

The school mean in Equation (2) is defined over the school’s student population, not over the students who happen to be sampled. In the simulation of Section 5, the population school mean is available by construction and enters the model directly. In the application, it must be estimated. Because students are sampled within schools with unequal probabilities, an unweighted mean of the sampled students would estimate a different quantity when within-school selection is informative. Therefore, the analysis uses the *weighted* school mean $\bar{x}_{.j} = \sum_{i \in j} w_{ij} x_{ij} / \sum_{i \in j} w_{ij}$, with weights w_{ij} defined below. The between-school coefficient is accordingly a gradient in this estimated school mean, a point Section 6.3 returns to.

Grouping all parameters $\psi = (\boldsymbol{\beta}, \tau^2, \sigma^2)$, the weighted likelihood of the analysis model on PV m is

$$\tilde{L}_m(\psi) = \prod_{j=1}^J \int \prod_{i \in j} f(\theta_{ij}^{(m)} \mid x_{ij}, \zeta_j; \boldsymbol{\beta}, \sigma^2)^{w_{ij}} dF(\zeta_j; \tau^2), \quad (3)$$

where f is the Gaussian density implied by Equation (2) given the school effect, F is the $N(0, \tau^2)$ distribution of the school effects, and w_{ij} is the final student weight normalized to sum to the sample size within country. Each student’s log-likelihood contribution is scaled by w_{ij} inside the random-intercept integral, and the school-level contributions carry no separate weight. This is the level-1 placement described in Section 3.

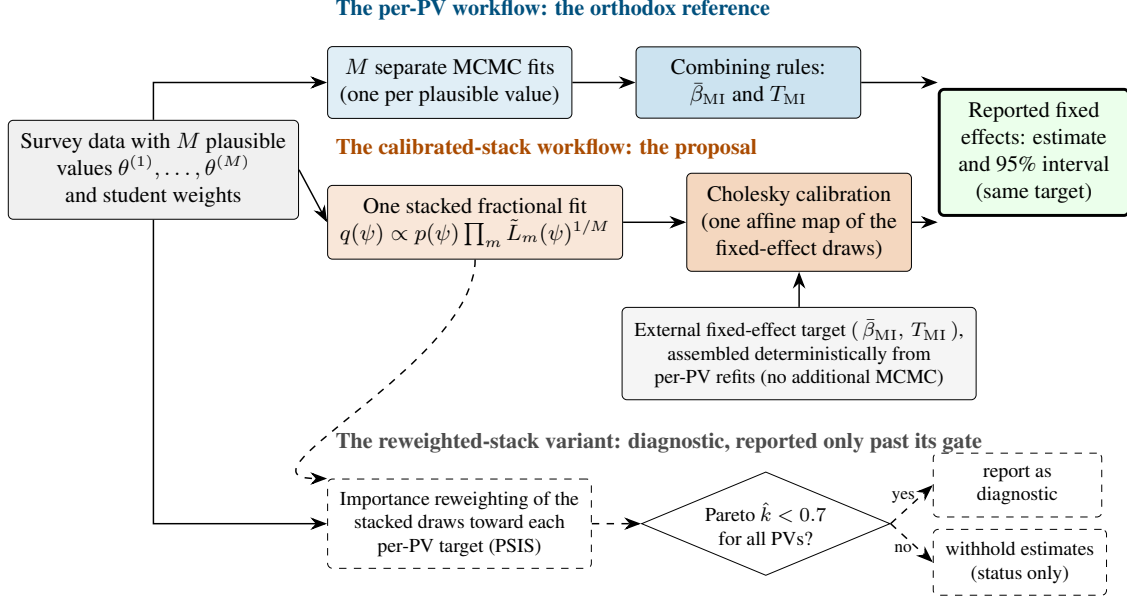


FIGURE 1. Three workflows for plausible-value inference in Bayesian multilevel models. The per-PV workflow (top) is the orthodox reference: M sets of MCMC results pooled by the combining rules. The calibrated-stack workflow (middle) is the proposal: one stacked fractional fit whose fixed-effect draws are calibrated, by a single affine map, to the externally assembled combining-rule target. Both workflows report the same target quantities. The reweighted-stack variant (bottom) recovers per-PV results from the same stacked draws by importance reweighting and is reported only when its Pareto- $\hat{k}$ reliability gate passes for every plausible value.

Figure 1 summarizes the three workflows this section develops, and Table 1 provides a reference of their key components.

4.2. The Per-PV Workflow. The per-PV workflow is the combining convention applied literally. For each m , an MCMC sampler is run for the posterior built on $\tilde{L}_m$, yielding a posterior mean $\hat{\beta}_m$ and posterior covariance $\hat{U}_m$; the M results are pooled through Equation (1), and intervals use the student- t reference with Barnard–Rubin degrees of freedom (Barnard & Rubin, 1999). Its use is uncontroversial, and it therefore defines the *target* that the other workflows must reproduce. Its cost is M sampler runs, M convergence audits, and M stored model objects for every model the analyst considers.

TABLE 1. The three workflows at a glance. M is the number of plausible values; $(\bar{\beta}_{\text{MI}}, T_{\text{MI}})$ is the combining-rule target of Equation (1).

	Per-PV workflow	Calibrated stack	Reweight stack
Role in this article	Orthodox reference	Proposal	Reliability-gated variant
MCMC fits per model	M	1	1 (reuses the stacked fit)
After the fit	Combining rules across M posteriors	One affine calibration to $(\bar{\beta}_{\text{MI}}, T_{\text{MI}})$	M importance reweightings, then combining rules
External input	None	Target assembled from deterministic per-PV refits	None
Convergence audits	M	1	1
Reportable when	Sampler diagnostics pass	Sampler diagnostics pass; target covariances positive definite	Sampler diagnostics pass and Pareto $\hat{k} < 0.7$ for all M PVs

The secondary model applied to stacked completions cannot achieve the target variance because the plausible values are draws from a posterior distribution over latent traits rather than realizations of a fixed value with added noise. The PV draws, in other words, are not equivalent to underlying data composed of student item responses. Unlike the setting in which the raw data are modeled directly, the variation across completions is not to be shrunk away but propagated, and the convention propagates it by modeling each PV draw separately and applying the combining rules.

The workflow developed next estimates the secondary analysis model once on the stacked completions and calibrates the posterior draws so that the reported estimator achieves the point and variance targets.

4.3. The Stacked Fractional Posterior. The stacked fractional PV posterior pools the M weighted likelihoods of Equation (3) in a single fit,

$$q(\psi) \propto p(\psi) \prod_{m=1}^M \tilde{L}_m(\psi)^{1/M}, \quad (4)$$

where $p(\psi)$ is the prior. The $1/M$ exponent makes the data enter as a geometric average of the M PV likelihoods rather than as M reuses of the same students, so

the stacked posterior carries one sample’s worth of information (Bissiri et al., 2016; Holmes & Walker, 2017). Operationally, Equation (4) is a single long-data model: the M PV columns are stacked into nM rows, with n the number of students, each row weighted by $1/M$ times its student’s normalized sampling weight, and one MCMC run samples q . The term *stacking* here refers to this fractional pooling of the PV likelihoods within one fit, not to the model-combination technique of the same name (Yao et al., 2018).

The rationale for stacking, rather than selecting a single PV or averaging the PVs into a composite outcome, is that the stacked fixed-effect center coincides exactly with the quantity the convention reports.

Theorem 4.1 (Stacked fixed-effect point identity). *Suppose (R1) all M PV analyses share the same analytic rows and cluster membership; (R2) the design matrix X is common across PVs; (R3) the marginal working covariance V of the outcome vector is common across PVs and held fixed; (R4) the PV weights are equal at $1/M$; (R5) the prior on β is flat and $X^\top V^{-1}X$ is invertible. Then the maximizer of the stacked objective in β equals the average of the per-PV generalized-least-squares estimators,*

$$\hat{\beta}^q = \arg \max_{\beta} \frac{1}{M} \sum_{m=1}^M \log \tilde{L}_m(\beta; V) = \frac{1}{M} \sum_{m=1}^M \hat{\beta}_m = \bar{\beta}_{\text{MI}},$$

exactly; and because the stacked objective is then quadratic in β , the q -marginal of β is Gaussian, so its mode and mean coincide.

The proof is one observation. At a fixed working covariance, each per-PV score is linear in $\theta^{(m)}$, so averaging the M normal equations gives the stacked normal equation, whose solution is the average of the per-PV solutions. The full proof appears in the online supplement (Appendix A.3), together with a numerical verification whose largest residual across a 24-cell grid is 4.4×10^{-16} ; Appendix A.4 delimits what the identity in Theorem 4.1 does and does not cover. The identity is a statement about the fixed-effect point under the stated regime. In production fits, where the variance components are estimated jointly, priors are weakly informative, and MCMC noise enters, the stacked center tracks the pooled center rather than equaling it by theorem, and the calibration step below makes the remaining difference immaterial: the reported center is set to the target directly.

4.4. Calibration of the Stacked Fractional Posterior. The stacked fit cannot supply the total variance: as Section 4.2 established, the between-PV component B

exists only across separate per-PV analyses. The calibrated-stack workflow therefore supplies the target to the fit, in two steps.

Assemble the target. The quantities in Equation (1) require per-PV estimates and covariances, but nothing requires those to come from the full posterior distribution (e.g. MCMC samples). Deterministic per-PV fits of the same model, by restricted maximum likelihood (REML) in our implementation (Bates et al., 2015), supply $\hat{\beta}_m$ and $\hat{U}_m$, and the combining rules do the rest, yielding a center $\bar{\beta}_{\text{MI}}$ and covariance T_{MI} without sampling, random seeds, or convergence considerations. The simulation in Section 5 shows this deterministic assembly tracking the Bayesian per-PV workflow closely across the design, which supports its use as the calibration target in practice.

The assembly is agnostic about the variance definition its inputs embody, and the workflow accepts both conventions in use. With model-based inputs, $\hat{U}_m$ is the REML covariance and the target is congenial to the analysis model; this is the default deployed in practice. When the reporting standard calls for a design-based variance instead, the same arithmetic accepts per-PV covariances built from the assessment’s replicate weights: PISA 2022 supplies 80 balanced-repeated-replication weights with Fay’s modification, each PV’s model is refit deterministically once per replicate weight, and the replicate-based covariance replaces $\hat{U}_m$ in Equation (1) (Judkins, 1990; OECD, 2024b). The calibrated stack then reports a design-based total variance from the same single MCMC fit. The replicate refits are deterministic and workflow independent; any workflow that reports a design-based variance incurs them, and they are computed once and reused across prior and sampler settings. The assembly routes, their costs, and the reason the main-text PISA analysis calibrates instead to the per-PV workflow’s own output are documented in the online supplement (Appendix C, with the replicate-weight variant in Appendix C.3).

Calibrate the draws. Let $\{\beta_s\}_{s=1}^S$ be the stacked fixed-effect draws with empirical mean $\bar{\beta}_{\text{raw}}$ and covariance Σ_{raw} . The Cholesky calibration correction (CCC) maps every draw through

$$\beta_s^{\text{cal}} = \bar{\beta}_{\text{MI}} + A(\beta_s - \bar{\beta}_{\text{raw}}), \quad A = L_{\text{target}} L_{\text{raw}}^{-1}, \quad (5)$$

where L_{target} and L_{raw} are the lower-triangular Cholesky factors of T_{MI} and Σ_{raw} . The map is deterministic, requiring one triangular solve and one matrix multiplication per draw with no iteration or tuning, and it satisfies three invariants by construction: the calibrated draws have mean exactly $\bar{\beta}_{\text{MI}}$; their covariance is exactly T_{MI} , to machine precision; and the map touches only the fixed-effect block, so all other draws

pass through unchanged (proofs and preconditions in the online supplement, Appendices B.2 and B.3). It is the multilevel, block-restricted form of the scale-and-shape correction that Williams and Savitsky (2021) developed for pseudo-posteriors under complex sampling, and its determinism distinguishes it from iterative importance-weighted moment matching (Jedhoff et al., 2025). The correction is a moment calibration: it guarantees agreement with the supplied target, and whether intervals constructed in this way attain nominal coverage is an empirical question that the simulation addresses.

The calibrated-stack workflow is a composition: one stacked fractional fit, one target assembly, one calibration. The analyst inspects a single set of convergence diagnostics and retains a single model object, and the reported quantities are those of the convention, with interval estimates in our implementation reported as normal-reference intervals formed from the calibrated mean and covariance. In the PISA application we calibrate to the per-PV workflow’s actual output, so that any difference between the two workflows is fully visible; in general practice the target is the deterministic assembly.

4.5. The Reweighted-Stack Variant. An alternative use of the stacked draws is to recover each per-PV posterior from them directly by importance reweighting and then to pool as usual, following the importance-reweighting approach of Jedhoff et al. (2025) with the stacked fit in the role of the proposal. For each m , the stacked draws serve as the proposal, and the target is the per-PV posterior; each draw is reweighted by the ratio of the target density to the proposal density.

In our implementation, the per-PV targets follow the survey’s declared two-stage weight decomposition, in the staged multilevel pseudolikelihood form of Rabe-Hesketh and Skrondal (2006). Writing w_j^{sch} and w_{ij}^{cond} for the school and conditional student factors of that decomposition, normalized so that their product is the analysis weight w_{ij} of Equation (3) (construction from the released weight variables in the online supplement, Appendix F.2), the target log-likelihood on PV m is

$$\ell_m^{\text{stg}}(\psi) = \sum_{j=1}^J w_j^{\text{sch}} \log \int \prod_{i \in j} f(\theta_{ij}^{(m)} \mid x_{ij}, \zeta_j; \beta, \sigma^2)^{w_{ij}^{\text{cond}}} dF(\zeta_j; \tau^2), \quad (6)$$

so the school factor scales each school’s integrated likelihood contribution and the conditional student factor enters within the integral. The log importance ratio for

PV m at draw ψ_s is, up to an additive constant,

$$\log w_{m,s} = \ell_m^{\text{stg}}(\psi_s) - \frac{1}{M} \sum_{m'=1}^M \log \tilde{L}_{m'}(\psi_s),$$

the staged target against the stacked proposal of Equation (4). Each ratio therefore bridges both the change of plausible value and a change of weight placement relative to the level-1 stacked proposal; the placement change is a deliberate feature of the variant’s design, which requires the reweighting to reach the survey’s declared two-stage target rather than an object the proposal already matches.

The raw ratios are stabilized by Pareto-smoothed importance sampling (PSIS; Vehtari et al., 2024, 2025), which also emits, for each reweighting, the tail-shape diagnostic $\hat{k}_m$ and an effective sample size; values of $\hat{k}$ above 0.7 mark the reweighting as unreliable. PSIS-weighted per-PV means and covariances then enter the combining rules of Equation (1).

The variant requires no external target; its reliability, however, is an empirical property of the importance weights that must be checked case by case rather than established in advance. Each reweighting uses a geometric mixture over M components as the proposal for a single component, and when the per-PV posteriors are well separated relative to their spread (a separation that increases as samples grow) the importance ratios acquire heavy tails. We therefore treat the variant as gated by construction: it is reported only when $\hat{k}_m < 0.7$ for every plausible value, and withheld otherwise, with no provision for reporting results under moderately elevated $\hat{k}$. Section 6.4 shows both branches of this rule operating on real data in two countries analyzed identically; construction details, provenance, and the full diagnostics are documented in the online supplement (Appendix D).

5. Simulation Study

5.1. Design. The simulation addresses two questions: whether, across the range of data structures a PISA-style analysis encounters, the calibrated stack delivers the same answers as the per-PV workflow it replaces, and whether those answers are accurate. The data-generating model is the two-level structure of Equation (2): a latent student outcome with school random intercepts, a covariate with within- and between-school variation, and informative two-stage sampling in which schools are selected with probability proportional to size. Plausible values are generated from the

latent outcome under a Gaussian measurement model with reliability ρ_{PV} . The generating fixed effects are $\beta^W = 0.40$ and $\beta^B = 0.60$, and both gradients are evaluated estimands: every fit records the within-school and the between-school coefficient.

Twenty-four conditions are organized in three strata (Table 2). Two design profiles mirror the United States and Korea reading settings: outcome intraclass correlation (ICC_y) 0.20 paired with covariate intraclass correlation (ICC_x) 0.15, and 0.45 paired with 0.30. A *mainstream* stratum of 12 conditions varies the number of schools $J \in \{50, 100, 200\}$ and the mean school size $\bar{n} \in \{10, 20, 30\}$ over eight of the nine cells of their grid at the 0.20 profile, runs the 0.45 profile at $J \in \{100, 200\}$ with $\bar{n} = 20$, and adds one covariate-ICC variation at each profile’s $J = 100$, $\bar{n} = 20$ anchor, at $R = 100$ replications. A *small-school stress* stratum runs the ninth cell, $J = 50$ with $\bar{n} = 20$, under both profiles at $R = 400$ to resolve behavior where school-level information is thinnest. A *PV sensitivity* stratum of 10 conditions varies the PV reliability $\rho_{PV} \in \{0.85, 0.90, 0.95\}$ and the PV count $M \in \{5, 10, 20\}$ around the 0.20 anchor, whose own setting is $\rho_{PV} = 0.90$, $M = 10$ (eight conditions), and adds the 0.45 anchor and an $\bar{n} = 10$ variant of it, at $R = 200$. Every replication runs four estimators on identical data: an oracle benchmark (the generating values, used as the coverage reference), the deterministic per-PV assembly (restricted-maximum-likelihood fits pooled by Equation (1); “per-PV REML” in the exhibits), the per-PV workflow with MCMC (“per-PV MCMC”), and the calibrated stack. Each estimator runs at the full replication count of its condition, for 4,000 fits per estimator across the design. The MCMC legs share a fixed configuration of four chains of 2,000 iterations with 1,000 warmup, and all fits executed on a single pinned machine and toolchain; the evidence tables behind every number, the run’s provenance, and its convergence diagnostics are documented in the online supplement (Appendices E.2 and E.3).

5.2. Results. The three estimators agree closely with one another and with the generating values on both gradients. Table 3 summarizes each stratum, and Figure 2 displays every condition. In the mainstream stratum, with 1,200 fits per estimator, the mean estimate of β^W is 0.4013 against the generating value 0.40 for all three estimators (bias 0.0013, Monte Carlo standard error 0.0008), and the mean estimate of β^B is 0.5969 against 0.60 (bias -0.0031 , Monte Carlo standard error 0.0040). The empirical coverage of nominal 95% intervals in that stratum is 0.973 for the per-PV MCMC workflow, 0.973 for the per-PV REML assembly, and 0.959 for the calibrated stack on β^W , and 0.957, 0.953, and 0.948 on β^B . The small-school stress and PV-sensitivity

TABLE 2. Simulation design: 24 conditions in three strata. All conditions share the two-level model of Equation (2), informative two-stage (probability-proportional-to-size) school sampling, generating fixed effects $\beta^W = 0.40$ and $\beta^B = 0.60$, and four estimators run on identical replication data. R is the number of replications per condition; the full condition-by-condition grid appears in the online supplement (Appendix E.1).

Stratum	Conditions	R	Varied factors
Mainstream	12	100	Schools $J \in \{50, 100, 200\}$ and mean school size $\bar{n} \in \{10, 20, 30\}$, over eight cells at the 0.20 profile; the 0.45 profile at $J \in \{100, 200\}$, $\bar{n} = 20$; one covariate-ICC variation at each profile’s anchor. Profiles (outcome/covariate ICC): 0.20/0.15 (United States reading) and 0.45/0.30 (Korea).
Small-school stress	2	400	Both profiles at $J = 50$, $\bar{n} = 20$, at high replication.
PV sensitivity	10	200	PV reliability $\rho_{PV} \in \{0.85, 0.90, 0.95\}$ and PV count $M \in \{5, 10, 20\}$ around the 0.20 anchor (eight conditions); the 0.45 anchor and an $\bar{n} = 10$ variant of it.

strata repeat the pattern. Across strata, estimators, and gradients, stratum-level absolute bias is at most 0.0013 for β^W and 0.0051 for β^B , in no case exceeding two Monte Carlo standard errors, and coverage lies between 0.945 and 0.986. At the condition level, the calibrated stack tracks the per-PV reference closely on both gradients; the per-condition results, with Wilson intervals (Wilson, 1927) reflecting each replication count, are tabulated in the online supplement (Appendix E.3).

Sampler behavior was unremarkable. Convergence diagnostics, re-extracted for two replications of every condition under the recorded seed schedule (which reproduces the production chains exactly), show all potential-scale-reduction statistics at or below 1.020 with no divergent transitions, and across all 4,000 replications the per-PV MCMC estimates differ from the deterministic REML estimates by at most 0.0007 for β^W and 0.011 for β^B (online supplement, Appendix E.2).

Two of these agreements carry the argument. The first is between the calibrated stack and the per-PV MCMC workflow: across intraclass correlations, school counts, school sizes, PV reliabilities, and PV counts, one calibrated fit reproduces what M fits produce, for the between-school gradient as well as the within-school gradient.

TABLE 3. Simulation results for the two fixed-effect gradients (generating values $\beta^W = 0.40$ and $\beta^B = 0.60$), by stratum and estimator. Bias is the mean estimate minus the generating value across all fits in the stratum; MC-SE is its Monte Carlo standard error; coverage is the empirical coverage of nominal 95% intervals. Every estimator runs at the full replication count of every condition. The oracle benchmark is exact by construction and is omitted; convergence diagnostics are reported in the online supplement (Appendix E.2).

Estimator	Within-school slope β^W			Between-school slope β^B		
	Bias	MC-SE	Coverage	Bias	MC-SE	Coverage
<i>Mainstream (12 conditions, $R = 100$; 1,200 fits per estimator)</i>						
Per-PV (MCMC)	0.0013	0.0008	0.973	-0.0031	0.0040	0.957
Per-PV (REML)	0.0013	0.0008	0.973	-0.0031	0.0040	0.953
Calibrated stack	0.0013	0.0008	0.959	-0.0031	0.0040	0.948
<i>Small-school stress (2 conditions, $R = 400$; 800 fits per estimator)</i>						
Per-PV (MCMC)	0.0005	0.0013	0.986	0.0044	0.0074	0.951
Per-PV (REML)	0.0005	0.0013	0.986	0.0045	0.0074	0.945
Calibrated stack	0.0005	0.0013	0.976	0.0051	0.0074	0.953
<i>PV sensitivity (10 conditions, $R = 200$; 2,000 fits per estimator)</i>						
Per-PV (MCMC)	-0.0005	0.0006	0.974	-0.0023	0.0030	0.959
Per-PV (REML)	-0.0005	0.0006	0.973	-0.0022	0.0030	0.954
Calibrated stack	-0.0005	0.0006	0.962	-0.0024	0.0030	0.955

The second is between the two per-PV routes themselves: the deterministic REML assembly and the Bayesian per-PV workflow move together across the design, which is what qualifies the deterministic assembly to serve as the calibration target in practice. On this evidence, an analyst who assembles the target from REML refits and calibrates one stacked fit to it obtains the same inference as an analyst who carries out the full M -fold Bayesian convention.

5.3. Scope. The simulation evidence pertains to Gaussian PV-generating designs of the kind described above, evaluated on the two fixed-effect gradients the workflows are built to report. Within that scope the design is broad: it brackets the intraclass correlations of the two application countries, spans a fourfold range of school counts and a threefold range of school sizes, and extends PV reliability and PV count well beyond their PISA values. Extending the evaluation to plausible values generated by operational item-response-theory scaling (Mislevy et al., 1992; OECD, 2024b) is a natural next step, and Section 7 returns to it. The PISA application of Section 6

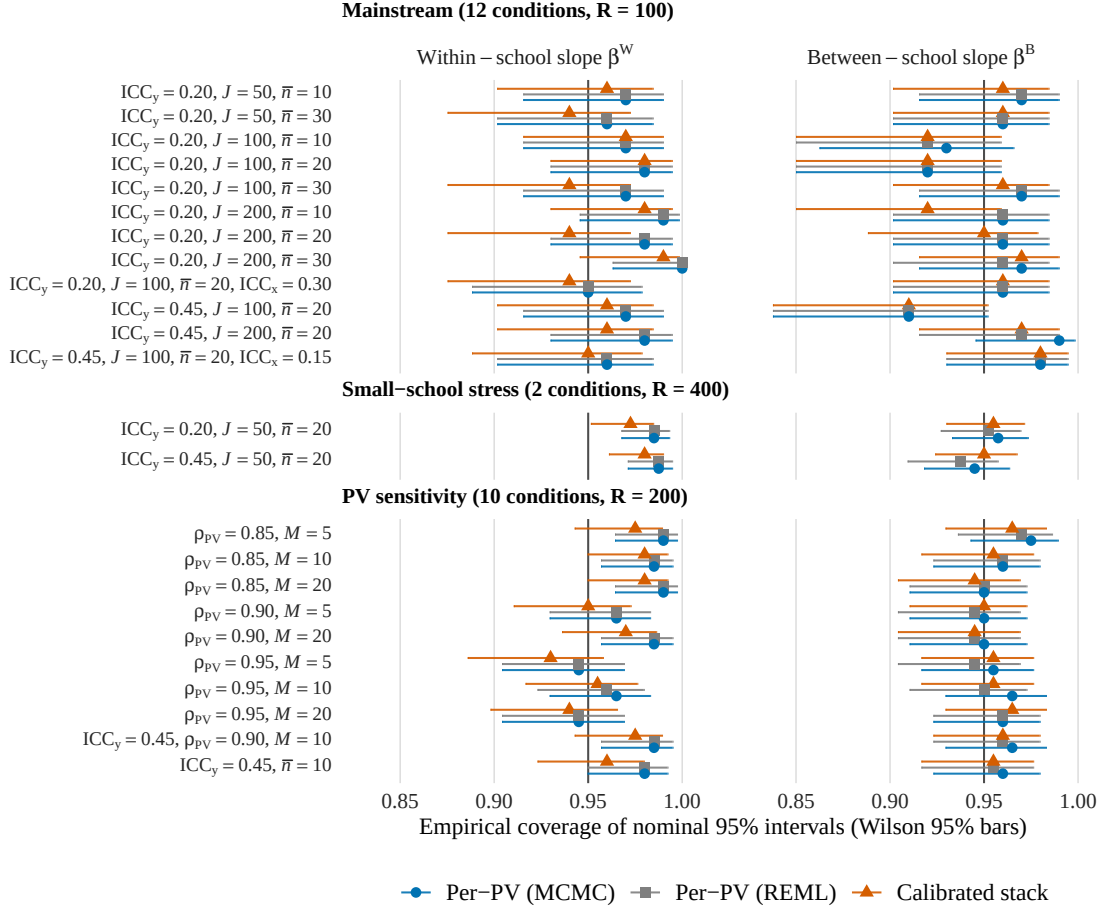


FIGURE 2. Empirical coverage of nominal 95% intervals for the within-school slope β^W (left, generating value 0.40) and the between-school slope β^B (right, 0.60), for the three estimators in all 24 conditions; panel headings give the stratum and its replication count. Labels give the design factors: outcome ICC (ICC_y ; 0.20 pairs with covariate ICC 0.15 and 0.45 with 0.30, except where ICC_x is shown), schools J , mean school size $\bar{n}$, and PV reliability ρ_{PV} and count M around the $ICC_y = 0.20$, $J = 100$, $\bar{n} = 20$ anchor. Horizontal bars are Wilson 95% intervals; the vertical line marks 0.95. Condition identifiers appear in the online supplement (Appendix E.1).

uses operational plausible values, but because its agreement is target-calibrated by construction, it demonstrates production implementation rather than an independent transfer of this simulation evidence to designs with IRT-generated plausible values.

6. The PISA 2022 Application

6.1. Objective. The application evaluates the workflow on data of operational scale and complexity. To make the comparison exact and auditable, we calibrate the stacked fit to the per-PV workflow’s own pooled output, so that the check is direct, namely whether one MCMC run, together with target assembly and one calibration, reproduces what ten MCMC runs produce. The resulting agreement is target-calibrated by design; the check demonstrates that the route operates on real data under a fixed analysis protocol, from analytic-sample definition and weights through target assembly, calibration, and diagnostics. The application also provides the reweighted-stack variant with its real-data test, and the two countries yield different outcomes.

6.2. Data and Model. We analyze PISA 2022 reading for the United States and Korea (OECD, 2024b): two education systems whose school structures differ in one relevant respect, in that the between-school share of outcome variance in the Korean sample is roughly twice that in the American sample. The outcome is the set of ten reading plausible values; the covariate is ESCS, decomposed into within- and between-school parts as in Equation (2). The school means are weighted by the same normalized final student weight that enters the likelihood, so that each school’s mean estimates its population ESCS mean under the within-school sampling design (Section 4.1; online supplement, Appendix F.1). The released samples contain 4,552 students in 154 schools (United States) and 6,454 students in 186 schools (Korea); removing students with missing ESCS (210 and 63) leaves analytic samples of 4,342 students in 150 schools and 6,391 students in 186 schools. The model fits weight each student’s likelihood contribution by the final student weight normalized to sum to the sample size within country, the level-1 placement declared in Section 3, as in Equation (3). The construction of the weights from the released weight variables, their two-stage decomposition into a school factor and a conditional student factor (whose product reconstructs the final student weight to numerical precision), and the sensitivity of a secondary between-country contrast to weight placement are documented in the online supplement (Appendices F.2 and F.4). Samplers ran in Stan (Carpenter et al., 2017): twenty per-PV fits for the per-PV workflow (two countries by ten PVs) and one stacked fractional fit per country for the calibrated stack, each stacked fit retaining 1,000 draws. Every fit passed its diagnostics without divergent

transitions; the largest potential-scale-reduction statistic was 1.016 across the twenty per-PV fits and 1.008 across the two stacked fits.

6.3. Agreement with the Per-PV Workflow. Table 4 and Figure 3 present the results. The calibrated stack reproduces the per-PV workflow on all four estimands: a within-school ESCS effect of 25.08 points per ESCS unit (SE 1.94) in the United States and 27.86 (1.98) in Korea, and a between-school effect of 69.74 (6.05) in the United States and 92.94 (6.54) in Korea, identical across the two workflows in estimate and standard error. Interval endpoints differ by less than 0.1 score point, because the per-PV workflow uses a student- t reference with Barnard–Rubin degrees of freedom (Barnard & Rubin, 1999) while the calibrated stack reports normal-reference intervals formed from the calibrated mean and covariance. The one-MCMC route therefore performs on operational data as designed. Table 4 is accordingly not offered as independent evidence that the two workflows estimate the same target; it verifies that the data cascade, weighting rule, target assembly, sampler output, and calibration operator jointly reproduce the per-PV target in a production PISA analysis. Substantively, both countries display the familiar pattern that the between-school ESCS gradient is several times the within-school gradient (Rabe-Hesketh & Skrondal, 2006). We present these values as context rather than as findings and draw no between-country comparison, for two reasons. First, the between-school contrast across countries is sensitive to the placement of the school-level weight, a sensitivity the online supplement examines directly (Appendix F.4). Second, the between-school coefficient is a gradient in the estimated weighted school mean of ESCS (Section 4.1) rather than a latent contextual effect corrected for school-mean measurement error. Both caveats reinforce the role of Table 4 as an implementation check rather than a cross-national comparison.

6.4. The Reweighted-Stack Variant and Its Reliability Gate. The reweighted-stack variant was built identically in both countries: the stacked draws served as the proposal, each of the ten per-PV reweightings was stabilized by PSIS, and the tail diagnostic $\hat{k}$ was recorded for every plausible value. Figure 4 displays the diagnostics. In the United States, all ten reweightings are usable, with $\hat{k}$ between 0.35 and 0.67, and the reweighted estimates lie close to the per-PV values: 25.03 (SE 1.93) within school and 68.75 (6.82) between schools, differences of -0.06 and -0.99 against the per-PV workflow (Table 5). In Korea, all ten reweightings fail, with $\hat{k}$ between 0.82 and 1.34 against the 0.7 threshold and PSIS effective sample sizes collapsing to between

TABLE 4. PISA 2022 reading: within-school and between-school ESCS effects under the per-PV workflow (ten MCMC fits per country) and the calibrated stack (one MCMC fit per country, calibrated to the per-PV target). Estimates are in reading score points per ESCS unit; intervals are 95%. Agreement is target-calibrated by design. Analytic samples: United States $n = 4,342$ students in $J = 150$ schools; Korea $n = 6,391$ in $J = 186$. The per-PV workflow uses a student- t reference with Barnard–Rubin degrees of freedom (Barnard & Rubin, 1999); the calibrated stack reports normal-reference intervals formed from the calibrated mean and covariance.

Country	ESCS effect	Per-PV workflow			Calibrated stack		
		Est.	SE	95% interval	Est.	SE	95% interval
United States	Within	25.08	1.94	[21.27, 28.89]	25.08	1.94	[21.29, 28.88]
United States	Between	69.74	6.05	[57.88, 81.60]	69.74	6.05	[57.89, 81.59]
Korea	Within	27.86	1.98	[23.91, 31.82]	27.86	1.98	[23.98, 31.74]
Korea	Between	92.94	6.54	[80.08, 105.79]	92.94	6.54	[80.11, 105.77]

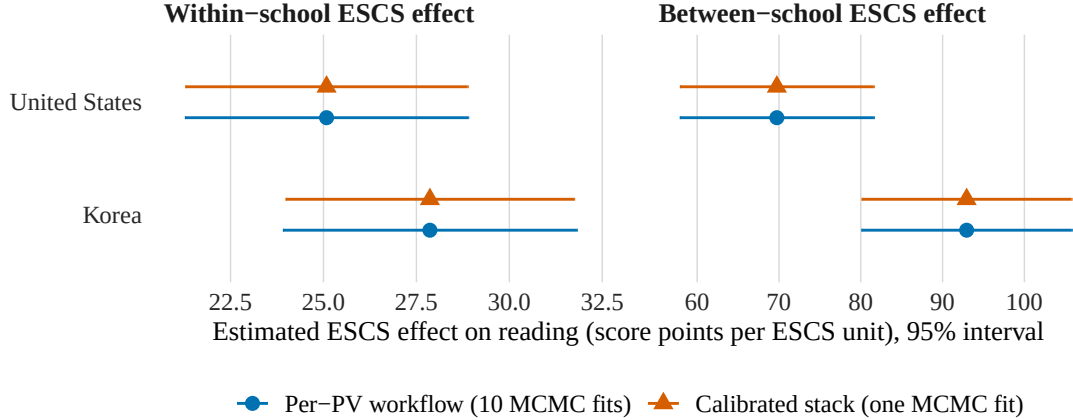


FIGURE 3. PISA 2022 reading: estimates and 95% intervals for the within-school and between-school ESCS effects under the per-PV workflow (ten MCMC fits per country) and the calibrated stack (one MCMC fit per country, calibrated to the per-PV target). The workflows coincide by construction of the calibration; the figure displays the implementation check.

5 and 30 of the 1,000 proposal draws. Following the gate-and-withhold rule fixed in advance, the Korean estimates are withheld and only the gate status is reported.

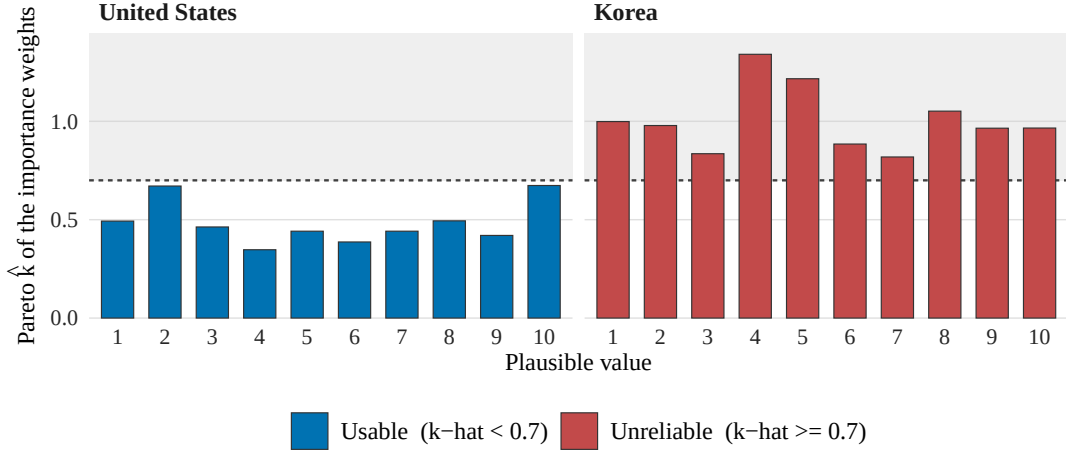


FIGURE 4. Pareto $\hat{k}$ of the importance weights for each of the ten reading plausible values (PV1READ–PV10READ), by country. The dashed line and shaded band mark the 0.7 reliability threshold. Every United States reweighting is usable; every Korean reweighting is unreliable, and the Korean estimates are therefore withheld.

TABLE 5. The reweighted-stack variant in the United States, where all ten plausible values pass the $\hat{k}$ gate. Differences and standard-error ratios are against the per-PV workflow and are computed from unrounded values; the reweighted per-PV targets follow the survey’s declared two-stage weight placement (online supplement, Appendix D). Korea’s estimates are withheld because every plausible value fails the gate (Figure 4).

ESCS effect	Estimate	SE	Diff. vs. per-PV	SE ratio
Within-school ESCS	25.03	1.93	-0.056	0.996
Between-school ESCS	68.75	6.82	-0.989	1.129

The contrast between the two countries admits a mechanical interpretation that is consistent with the diagnostics and with the structural risk identified in Section 4.5. An importance ratio is well behaved when its target lies within the effective support of its proposal (Vehtari et al., 2024); here the proposal is a geometric average of ten per-PV posteriors, and each target is one of the components. On this account, the ten per-PV posteriors in the American sample lie close enough together that their geometric average is an adequate proposal for any one of them. The Korean sample is larger and more strongly clustered, which concentrates each per-PV posterior relative to the between-PV separation, so that the geometric average lies between

the components and the importance ratios that must carry draws from the average to a single component acquire heavy tails. The collapse of the effective sample sizes is consistent with this account. We did not separately measure the geometry of the per-PV posteriors, however, so the account should be read as an interpretation of the PSIS diagnostics rather than as an established mechanism. The staged per-PV targets of Equation (6) also require the ratios to bridge a change of weight placement as well as a change of plausible value; this is an additional gap for the proposal to span, although the United States data accommodate it without difficulty.

This behavior underlies the article’s recommendation to calibrate rather than reweight. Reweighting makes every estimate contingent on a diagnostic that operational data can fail entirely, and in this application the failure arose in the larger and more strongly clustered of the two samples, the kind of data that large-scale assessment routinely produces. Calibration involves no importance weights and is therefore not exposed to this tail-weight failure mode, although its own requirements remain those of Section 4.4: a trusted target and a stacked fit whose diagnostics pass. The Korean dataset on which the reweighted stack fails presents no difficulty for the calibrated stack (Table 4). Where the gate passes, as in the United States, the reweighted stack provides a useful cross-check on the calibrated results; where it fails, the estimates are withheld and only the gate status is reported.

7. Discussion

The calibrated stack gives the applied Bayesian analyst the convention’s own report from a single sampler run. The simulation shows the workflow accurate and well calibrated for both the within- and between-school gradients across the data structures PISA-style analyses encounter, in step with the per-PV convention across the design; the PISA application shows it performing on operational data in two countries. The workflow changes the structure of the analyst’s work. Model comparison, sensitivity analysis, and country-by-country replication each require one sampler fit, together with the corresponding deterministic target assembly, rather than M sampler fits; there is one set of convergence diagnostics to defend in review rather than M ; and, for a given model, the target assembly (the deterministic per-PV arithmetic that defines the reportable quantities) is computed once, without sampling, and reused across prior and sampler-setting choices. The gain is a single fit, a single convergence audit, and a single calibrated draw cloud per model; we make no claim about

processor time, which depends on implementation details that the topology does not control.

For a single, final, high-stakes model, the per-PV convention remains a sound default, and nothing here argues against it. The calibrated stack is intended for the remainder of the analysis lifecycle (model search, robustness checks, and multi-country replication) where the M -fold topology imposes a cost on every decision. Its requirements are the ones this article makes explicit, namely a target assembled from per-PV fits the analyst trusts and a stacked fit whose diagnostics pass. The calibration then transfers the target exactly. It cannot improve a poorly assembled target, and its usefulness lies in reproducing the convention from a single fit rather than in improving on it.

The reweighted stack should be used only when its gate passes. Where every plausible value clears $\hat{k} < 0.7$, it provides a cross-check, computed from the same draws, that does not reuse the calibration target, and in the United States it tracked the per-PV workflow to within a fraction of a standard error. The Korean analysis shows that the route can fail entirely on data that present no other difficulty. There, every plausible value was unreliable, with effective sample sizes as low as single digits, and the failure occurred in a large, strongly clustered sample of the kind that large-scale assessments produce. Analysts who adopt the variant should commit to the gate-and-withhold rule before examining results. Adaptive refinements of the importance-sampling step (Paananen et al., 2021) may extend the variant’s usable range, but they do not remove the diagnostic obligation.

Four limitations bound the scope of this article. First, the reportable quantities are fixed effects. The stacked fit estimates the variance components, and the calibration deliberately leaves their draws untouched, but we do not report them as inferential quantities, because the point identity does not extend to them and no external variance-component target is assembled here. Extending the calibration idea to variance components is the natural next problem. Second, the simulation generates plausible values from a Gaussian measurement model. Operational PVs come from item-response-theory scaling with a latent regression, and evaluating the workflow under an operational-style generator, with the congeniality questions that accompany it (Jehangir & Fox, 2025; Meng, 1994), remains for future work. Third, the model fits place the normalized final student weight at level 1, a declared choice suited to the student-population fixed effects reported here; other placements answer subtly different population questions (Asparouhov, 2006; Pfeffermann et al., 1998), and the

supplement’s sensitivity analysis (Appendix F.4) shows a secondary between-country contrast moving materially with that choice. Fourth, the empirical demonstration covers one model family in two countries; generalization across assessments and model classes will require accumulated applications.

The workflow’s structure is not specific to PISA. Other assessments that release plausible values, among them TIMSS, PIRLS, NAEP, and PIAAC, prescribe combining rules of the same multiple-imputation form for secondary analysis (Jewsbury et al., 2024; Wu, 2005), and the same division of labor applies, with a deterministic target assembly, one stacked fit, and one exact calibration; each deployment requires its own checks of PV generation, weighting, and target assembly. Within the boundaries stated above, the calibrated stack returns the convention’s report at the cost of one MCMC fit, and the reweighted variant, where its gate passes, supplies a cross-check from the same draws.

8. Software and Reproducibility

The three workflows are implemented in the open-source R package `pvstackr`. The package provides a declared-design interface in which the PV columns, weights, and replicate-weight structure are recorded before any fitting occurs, the deterministic target assembly of Section 4.4 in both its model-based and replicate-weight forms, the calibration operator with machine-checkable invariants, and the $\hat{k}$ gate-and-withhold rule for the reweighted variant, implemented as package behavior rather than left to user discretion. Analyses used R (R Core Team, 2026), Stan through the `cmdstanr` interface (Carpenter et al., 2017; Gabry et al., 2024), `lme4` for the deterministic per-PV refits (Bates et al., 2015), and the `loo` implementation of Pareto-smoothed importance sampling (Vehtari et al., 2024, 2025). Every number reported in this article traces to an archived evidence table generated under a recorded seed schedule; the online supplement documents the package interface, the archive and seeds, and the mapping between the workflow names used here and the labels in the archived tables (Appendix G). A public replication package bundles the frozen analysis engine, the shipped evidence, an automated verification suite that re-checks every reported number, and a rendered reproduction guide, and regenerates every table and figure in this article from included evidence; the repository and guide are listed in the reproducibility note at the end of the article.

Software and reproducibility. The three workflows are implemented in the open-source R package `pvstackr` (<https://github.com/joonho112/pvstackr>; documentation

at <https://joonho112.github.io/pvstackr/>), which draws posterior samples through Stan via the `cmdstanr` interface. A public replication package for this article (the frozen analysis engine, the shipped evidence, an automated verification suite that re-checks every reported number, and a rendered reproduction guide) is available at <https://github.com/joonho112/pvstackr-replication> (companion guide at <https://joonho112.github.io/pvstackr-replication/>); every table and figure in [Sections 5](#) and [6](#) and the online supplement regenerates from its included evidence.

Data availability. PISA 2022 data are publicly available from the OECD (<https://www.oecd.org/pisa/data/>). Under the current PISA-specific access terms the unit-record files may not be redistributed by a third party, so the replication package supplies the data-build scripts and an authorized-access contract rather than the student and school records themselves; the simulation is fully self-contained and regenerates its data from seeds.

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Online Supplement

One Markov Chain Monte Carlo Fit for Many Plausible Values: A Calibrated Stacked Posterior Workflow for Bayesian Multilevel Models of Large-Scale Assessment Data

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Appendix A. Notation, Setup, and the Stacked Fixed-Effect Point Identity

This appendix collects the notation used throughout the article and the supplement, states the stacked fixed-effect point identity (Theorem [A.1](#), Theorem 1 of the main text) with its conditions, gives the proof, and delimits what the identity does and does not cover.

A.1. Notation. [Table A.1](#) collects the symbols used in the main text and in this supplement.

TABLE A.1. Notation used in the main text and supplement.

Symbol	Meaning
$i = 1, \dots, n_j; j = 1, \dots, J$	Students within schools
$m = 1, \dots, M$	Plausible values (PISA: $M = 10$)
$\theta_{ij}^{(m)}$	m th plausible value for student i in school j
$x_{ij}, \bar{x}_{.j}$	Student covariate (ESCS) and its weighted school mean
$\beta = (\beta_0, \beta^W, \beta^B)^\top$	Fixed effects: intercept, within-school, between-school
τ^2, σ^2	School-intercept and residual variance components
$\psi = (\beta, \tau^2, \sigma^2)$	Full parameter
w_{ij}	Sampling weight, normalized to sum to the sample size within country
$\tilde{L}_m(\psi)$	Weighted likelihood of the analysis model on PV m , Equation (3) of the main text
$q(\psi)$	Stacked fractional PV posterior, Equation (4) of the main text
$\hat{\beta}_m, \hat{U}_m$	Per-PV estimate and within-imputation covariance
$\hat{\beta}_{\text{MI}}, B, \bar{U}, T_{\text{MI}}$	Combining-rule quantities, Equation (1) of the main text
$c, \Sigma_{\text{target}}$	Supplied calibration center and covariance target
$L_{\text{raw}}, L_{\text{target}}$	Lower-triangular Cholesky factors
$\hat{k}$	Pareto shape diagnostic of PSIS
V	Marginal working covariance of the outcome vector
X	Fixed-effect design matrix (common across PVs)

A.2. Setup. Stack the outcome for PV m into the vector $\theta^{(m)} \in \mathbb{R}^n$ ($n = \sum_j n_j$), and let $X \in \mathbb{R}^{n \times p}$ be the fixed-effect design matrix, identical across PVs because the covariates, analytic rows, and cluster membership do not vary with m . Under the two-level Gaussian model of Equation (2) of the main text, the marginal distribution of $\theta^{(m)}$ given β at fixed variance components is $N(X\beta, V)$, where $V = V(\tau^2, \sigma^2)$ is block diagonal over schools with blocks $\sigma^2 I_{n_j} + \tau^2 \mathbf{1}_{n_j} \mathbf{1}_{n_j}^\top$; sampling weights enter by scaling each student’s log-likelihood contribution. For the theorem, the only property used is that all M analyses share one common symmetric positive-definite working precision, denoted P , which may be the unweighted V^{-1} or a precision constructed from a declared weight placement; every step below uses only the common quadratic form $X^\top P X$ and a score linear in $\theta^{(m)}$. To keep the notation close to the unweighted case we write V^{-1} throughout, and substituting any common SPD P changes no step. Production weighted random-intercept likelihoods, in which the variance components are estimated jointly rather than held fixed, are outside the theorem and are treated empirically (main text, Section 4.3). Write $\ell_m(\beta; V) = \log \tilde{L}_m(\beta; V)$ for the weighted Gaussian log-likelihood of PV m at the fixed working covariance.

A.3. The Identity and Its Proof.

Theorem A.1 (Stacked fixed-effect point identity). *Assume (R1) all M PV analyses share the same analytic rows and cluster membership; (R2) the design matrix X is common across PVs; (R3) the working covariance V is common across PVs and held fixed; (R4) the PV weights are equal at $1/M$; (R5) the prior on β is flat and $X^\top V^{-1}X$ (equivalently, with any common SPD working precision P in place of V^{-1}) is invertible. Then*

$$\hat{\beta}^q = \arg \max_{\beta} \frac{1}{M} \sum_{m=1}^M \ell_m(\beta; V) = \frac{1}{M} \sum_{m=1}^M \hat{\beta}_m = \bar{\beta}_{\text{MI}},$$

where $\hat{\beta}_m = \arg \max_{\beta} \ell_m(\beta; V)$ is the per-PV generalized-least-squares (GLS) estimator. Moreover, under (R3) and the flat prior of (R5) the q -marginal of β is Gaussian, so its mode and mean coincide: $\hat{\beta}^q = \mathbb{E}_q[\beta]$.

Proof. At fixed V , each per-PV log-likelihood is, up to an additive constant not involving β ,

$$\ell_m(\beta; V) = -\frac{1}{2} (\theta^{(m)} - X\beta)^\top V^{-1} (\theta^{(m)} - X\beta),$$

a quadratic in β whose score is linear in $\theta^{(m)}$:

$$\nabla_{\beta} \ell_m(\beta; V) = X^\top V^{-1} (\theta^{(m)} - X\beta).$$

Setting the score to zero gives the per-PV normal equation $X^\top V^{-1}X \hat{\beta}_m = X^\top V^{-1}\theta^{(m)}$, whence $\hat{\beta}_m = (X^\top V^{-1}X)^{-1}X^\top V^{-1}\theta^{(m)}$ by (R5). The stacked objective is the equally weighted average $M^{-1} \sum_m \ell_m(\beta; V)$ by (R4), and under the flat prior of (R5) the prior contributes no β -gradient. Its score is the average of the per-PV scores,

$$\frac{1}{M} \sum_{m=1}^M X^\top V^{-1} (\theta^{(m)} - X\beta) = X^\top V^{-1} (\bar{\theta} - X\beta), \quad \bar{\theta} = \frac{1}{M} \sum_{m=1}^M \theta^{(m)},$$

using (R1)–(R3), so the stacked normal equation is $X^\top V^{-1}X \hat{\beta}^q = X^\top V^{-1}\bar{\theta}$, and

$$\hat{\beta}^q = (X^\top V^{-1}X)^{-1}X^\top V^{-1}\bar{\theta} = \frac{1}{M} \sum_{m=1}^M (X^\top V^{-1}X)^{-1}X^\top V^{-1}\theta^{(m)} = \frac{1}{M} \sum_{m=1}^M \hat{\beta}_m.$$

The averaging and the GLS map commute because the map is linear in θ . Finally, under (R3) and the flat prior of (R5) the stacked objective is a fixed quadratic in β , so the q -marginal of β is Gaussian and its mode equals its mean. $\square$

In implementation the identity is verified numerically rather than assumed. Across a 24-cell grid of test configurations varying M , cluster counts, cluster sizes, and intraclass correlations, the maximum absolute difference between the stacked GLS solution and the averaged per-PV solutions is 4.4×10^{-16} , at machine-epsilon level (the verification grid is part of the reproducibility archive, [Section G](#)).

A.4. Scope Discipline. Four boundaries, stated in or implied by the main text and collected here for completeness, govern every use of [Theorem A.1](#) in the article.

- (1) *Point identity only.* The theorem equates maximizers (and, under flatness, posterior means). It makes no statement about posterior variances, interval widths, or frequentist coverage. All variance statements in the article run through the calibration operator ([Section B](#)) and are evaluated empirically ([Section E](#)).
- (2) *Fixed effects only.* The per-PV profile score for (τ^2, σ^2) carries $\log \det V$ and quadratic terms, both nonlinear in $\theta^{(m)}$; averaging per-PV scores does not commute with the estimator map, and the stacked variance-component estimates can differ from per-PV pooled ones. This asymmetry is a structural reason the calibration is restricted to the fixed-effect block.
- (3) *Not a production theorem.* In production fits, V is estimated jointly with β (violating R3), priors are weakly informative (violating the flat-prior part of R5), and per-PV MCMC noise enters. Agreement between the stacked center and the pooled center in production is an empirical quantity to be measured and reported, never asserted as a corollary of the theorem. The production workflow therefore calibrates the center as well as the covariance ([Section B](#)).
- (4) *Not inherited by reweighting.* The reweighted-stack variant’s PSIS-weighted per-PV centers are simulation-consistent approximations, not exact quantities; the identity says nothing about them.

A final distinction is worth preserving. The theorem concerns the estimator; a separate, trivial *implementation identity* (that stacking M equally weighted copies of the data rows reproduces the same objective as averaging the M log-likelihoods) is a bookkeeping fact about the long-data representation. Verifying the second does not establish the first; the archive verifies both separately.

Appendix B. The Cholesky Calibration Correction

This appendix details the calibration operator of Section 4.4 of the main text: algorithm, invariants, preconditions, failure modes, and the binding of the operator to each workflow.

B.1. Algorithm. Inputs: fixed-effect draws $\{\beta_s\}_{s=1}^S$ from the stacked fractional fit; a supplied center $c \in \mathbb{R}^p$; a supplied symmetric positive-definite (SPD) covariance target $\Sigma_{\text{target}} \in \mathbb{R}^{p \times p}$.

- (1) Compute the empirical mean $\bar{\beta}_{\text{raw}}$ and covariance Σ_{raw} of the draws.
- (2) Verify preconditions: Σ_{raw} and Σ_{target} SPD (Cholesky succeeds); dimensions match; draw count S is much larger than p .
- (3) Factor $\Sigma_{\text{raw}} = L_{\text{raw}} L_{\text{raw}}^\top$ and $\Sigma_{\text{target}} = L_{\text{target}} L_{\text{target}}^\top$ with lower-triangular factors. (In R, `chol()` returns the upper factor; the implementation uses its transpose. Mixing triangular conventions produces a valid covariance but a different, unintended rotation of the draw cloud, so one convention must be used throughout.)
- (4) Form $A = L_{\text{target}} L_{\text{raw}}^{-1}$ by one triangular solve, and map every draw: $\beta_s^{\text{cal}} = c + A(\beta_s - \bar{\beta}_{\text{raw}})$.
- (5) Reassemble the posterior draws: calibrated fixed-effect block, original (untouched) variance-component draws.
- (6) Emit diagnostics: the achieved mean and covariance of the calibrated cloud, their residuals against $(c, \Sigma_{\text{target}})$, and the condition numbers of both factors.

The map is deterministic (no iteration, no tuning), costs one triangular solve plus one matrix multiplication per draw, and is invertible, so the raw cloud can always be recovered.

B.2. Invariants.

Proposition B.1 (Exact moment calibration). *Let $\beta_s^{\text{cal}} = c + A(\beta_s - \bar{\beta}_{\text{raw}})$ with $A = L_{\text{target}} L_{\text{raw}}^{-1}$ as above. Then (i) the empirical mean of $\{\beta_s^{\text{cal}}\}$ is exactly c ; (ii) the empirical covariance of $\{\beta_s^{\text{cal}}\}$ is exactly Σ_{target} ; (iii) draws of parameters outside the fixed-effect block are unchanged.*

Proof. (i) Averaging the map gives $c + A(\bar{\beta}_{\text{raw}} - \bar{\beta}_{\text{raw}}) = c$. (ii) The empirical covariance of the mapped cloud is $A \Sigma_{\text{raw}} A^\top = L_{\text{target}} L_{\text{raw}}^{-1} (L_{\text{raw}} L_{\text{raw}}^\top) L_{\text{raw}}^{-\top} L_{\text{target}}^\top = L_{\text{target}} L_{\text{target}}^\top = \Sigma_{\text{target}}$. (iii) The map is defined on the fixed-effect coordinates only. $\square$

Both identities hold to machine precision in floating point (residuals of order 10^{-16} in the implementation's unit tests). Because the empirical moments, rather than population moments, are matched, the calibration uses the realized draw cloud as it stands; MCMC noise in Σ_{raw} affects only the rotation A , not the achieved target moments.

B.3. Preconditions, Failure Modes, and What CCC Cannot Do. The operator halts with an error, rather than degrading silently, when its preconditions fail. Σ_{raw} can be rank-deficient when draws are too few or a fixed effect is unidentified; Σ_{target} can fail SPD when assembled from a degenerate between-imputation component (for example, M too small relative to p) or from inconsistent per-PV inputs. The implementation surfaces these as hard errors with the offending eigenvalues; a nearest positive-semidefinite repair (Higham, 1988) is available as an explicitly logged fallback but was not needed in any analysis reported in the article.

The operator deliberately does not do three things. First, it does not correct a wrong center: the calibrated mean is whatever center is supplied, so target-assembly errors pass through untouched. Second, it does not calibrate the variance components: the point identity does not license their center (Section A.4), no external variance-component target is assembled in this article, and an affine map applied to draws of a positivity-constrained parameter near its boundary can produce inadmissible values; their draws pass through unchanged and are not reported as inferential quantities. Third, it does not certify coverage: the invariants of Proposition B.1 are moment identities, and whether intervals built from the calibrated draws cover at nominal rates is an empirical question, answered descriptively by the simulation (Section E).

B.4. Workflow Binding. The operator is generic in $(c, \Sigma_{\text{target}})$; the workflows differ in what they bind to it.

TABLE B.1. Binding of the calibration operator across workflows.

Workflow	Calibration calls	Center c	Target Σ_{target}
Per-PV workflow	none required ^a	—	—
Calibrated stack	1 (pooled)	$\bar{\beta}_{\text{MI}}$	T_{MI}
Rewighted-stack variant	M (per PV, PSIS-weighted moments)	$\hat{\beta}_m^{\text{PSIS}}$	per-PV target

^aThe per-PV workflow produces the combining-rule quantities directly; when a design-based covariance target is wanted instead of the model-based one, the same operator applies per PV (Section C).

In the PISA application (Section 6 of the main text), the calibrated stack binds $(c, \Sigma_{\text{target}})$ to the per-PV workflow’s pooled output, which is what makes the reported agreement between the two workflows exact by construction; the agreement is an implementation check of the one-MCMC route, not independent validation. In deployed practice the target is assembled without MCMC (Section C).

Appendix C. Assembling the External Fixed-Effect Target

The calibrated stack consumes a supplied center and covariance target $(\bar{\beta}_{\text{MI}}, T_{\text{MI}})$. This appendix documents the assembly routes: the generic combining-rule assembly, the deterministic frequentist route used in deployed practice, the design-based replicate-weight variant, and the structural reason no single posterior can substitute for the assembly.

C.1. The Combining-Rule Assembly. Given per-PV fixed-effect estimates $\hat{\beta}_m$ and within-imputation covariances $\hat{U}_m$, the multiple-imputation quantities are assembled exactly as in Equation (1) of the main text (Rubin, 1987):

$$\bar{\beta}_{\text{MI}} = \frac{1}{M} \sum_{m=1}^M \hat{\beta}_m, \quad \bar{U} = \frac{1}{M} \sum_{m=1}^M \hat{U}_m, \quad B = \frac{1}{M-1} \sum_{m=1}^M (\hat{\beta}_m - \bar{\beta}_{\text{MI}})(\hat{\beta}_m - \bar{\beta}_{\text{MI}})^\top, \quad (\text{C.1})$$

with $T_{\text{MI}} = \bar{U} + (1 + 1/M)B$. For scalar interval reporting the implementation also emits the per-coordinate Barnard–Rubin degrees of freedom (Barnard & Rubin, 1999) and the fraction of missing information. The assembly is agnostic about where $(\hat{\beta}_m, \hat{U}_m)$ come from; the routes below differ only in that respect.

C.2. The Deterministic Frequentist Route. In deployed practice the per-PV inputs are restricted-maximum-likelihood (REML) fits of the same two-level model (Bates et al., 2015), requiring M deterministic optimizations and involving no MCMC, convergence audits, or random seeds. $\hat{\beta}_m$ is the REML fixed-effect estimate and $\hat{U}_m$ its model-based covariance. Two properties make this route the recommended default. First, it is congenial to the analysis model: the same likelihood underlies the sampler and the target, so the target estimates the same population quantity. Second, it is empirically aligned with the orthodox Bayesian reference: in the simulation (Appendix E), the frequentist per-PV leg and the Bayesian per-PV leg track each other closely in point recovery and coverage across the Gaussian design, which is the evidence that calibrating to the frequentist-assembled target directs the one-MCMC workflow to the quantity the per-PV workflow would have reported.

C.3. The Design-Based Replicate-Weight Variant. When a design-based (rather than model-based) covariance is wanted, the same assembly accepts per-PV covariances built from the assessment’s replicate weights. PISA 2022 supplies $R = 80$ balanced-repeated-replication weights with Fay’s modification ($k = 0.5$) (Judkins, 1990; OECD, 2024b). For PV m , the model is refit deterministically once per

replicate weight, yielding $\hat{\beta}_m^{(r)}$, and

$$\hat{U}_m^{\text{BRR}} = a_d \sum_{r=1}^R (\hat{\beta}_m^{(r)} - \hat{\beta}_m) (\hat{\beta}_m^{(r)} - \hat{\beta}_m)^\top, \quad a_d = \frac{1}{R(1-k)^2} = \frac{1}{80 \times 0.25} = 0.05. \quad (\text{C.2})$$

Substituting $\hat{U}_m^{\text{BRR}}$ for $\hat{U}_m$ in Equation (C.1) yields a design-based T_{MI} . The full loop costs $M(R+1)$ deterministic refits (810 per country at PISA scale); this is a shared, workflow-independent computation that any of the three workflows would equally require if a design-based target is the reporting standard. Two caveats are recorded for completeness. First, PISA releases student-level replicate weights only (OECD, 2024a), so school-level replicate allocation is not identified from the public files. Second, this design-based variant is not the target used in the main-text PISA analysis, which calibrates to the per-PV workflow model-based target to keep the implementation check exact (Section B.4).

C.4. Why a Single Posterior Cannot Substitute. Within the orthodox PV-as-completed-data combining framework, the between-imputation component B in Equation (C.1) is a covariance of M estimates, each a functional of a different PV vector. A single posterior, stacked or otherwise, conditions on one realized data object and cannot manufacture the cross-PV dispersion of estimators from its own draws; the stacked posterior’s fixed-effect covariance approximates a one-sample information quantity, which corresponds to the $\bar{U}$ part only. Concretely, if one applies a replicate-weight loop to the stacked fit itself, the resulting covariance estimates $\bar{U}$, not T_{MI} ; it is missing exactly the $(1 + 1/M)B$ term, and the shortfall grows with the fraction of missing information. This is not a defect of the stacked fit; it is the structural reason the workflow assembles the target externally and treats calibration, rather than reading variances off the sampler, as the reporting step (Little, 2011; Meng, 1994).

Appendix D. The Reweighted-Stack Variant: Construction, Diagnostics, and Full PISA Results

The reweighted-stack variant is the diagnostic-gated reweighting extension of Section 4.5 of the main text. Per the reporting rule fixed in advance, its construction, full diagnostics, and provenance are documented here, and the main text reports the headline results (Section 6.4); it is not a primary estimator in this article.

D.1. Construction. Let $\{\psi_s\}_{s=1}^S$ be draws from the stacked fractional posterior q (the same draws the calibrated stack consumes; $S = 1,000$ in the PISA application). Under the frozen estimand contract, the per-PV target of the reweighting is the declared SNS-01 *staged marginal* pseudo-posterior (contract field `pipeline_b_numerator_source = sns01_staged_full_sample`), whose log-likelihood on PV m is Equation (6) of the main text, restated here in the notation of this supplement. Writing w_j^{sch} and w_{ij}^{cond} for the two weight factors of Equation (F.1),

$$\ell_m^{\text{stg}}(\psi) = \sum_{j=1}^J w_j^{\text{sch}} \log \int \prod_{i \in j} f(\theta_{ij}^{(m)} \mid \zeta_j; \psi)^{w_{ij}^{\text{cond}}} dF(\zeta_j; \psi), \quad (\text{D.1})$$

so the conditional student factors scale each student’s contribution inside the random-intercept integral while the school factor multiplies the school’s integrated log contribution, the staged multilevel pseudolikelihood form of Rabe-Hesketh and Skrondal (2006). This staged placement differs from the level-1 placement used by the per-PV and calibrated-stack fits (Section F.2); it is a deliberate feature of the diagnostic’s frozen design, enforced by a firewall that fails hard if final or replicate weights are supplied as the numerator.

For each PV m , the log importance ratio of this target against the stacked proposal is, up to a constant,

$$\log w_{m,s} = \ell_m^{\text{stg}}(\psi_s) - \frac{1}{M} \sum_{m'=1}^M \log \tilde{L}_{m'}(\psi_s), \quad (\text{D.2})$$

where $\tilde{L}_{m'}$ is the level-1-weighted likelihood that enters the stacked fractional proposal (Equations (3) and (4) of the main text). Each ratio therefore reflects both the change of plausible value and the difference in weight placement between the staged target and the level-1 proposal, and the PSIS diagnostic below gates the reliability of the reweighting toward that declared staged target.

The raw ratios are stabilized by Pareto-smoothed importance sampling (PSIS; Vehtari et al., 2024, 2025), which also emits the tail-shape diagnostic $\hat{k}_m$ and an effective sample size. PSIS-weighted per-PV centers $\hat{\beta}_m^{\text{PSIS}} = \sum_s \tilde{w}_{m,s} \beta_s$ and covariances then enter the combining-rule assembly of Equation (C.1). The reliability gate is pre-committed: numeric estimates from the reweighted-stack variant are reportable only if $\hat{k}_m < 0.7$ for every PV; otherwise the estimates are withheld and only the gate status is reported. The gate is one-directional by design: there is no provision for reporting results with moderately elevated $\hat{k}$. The 0.7 ceiling is the fixed form of

the PSIS threshold; Vehtari et al. (2024) also give a sample-size-dependent form, $\min\{1 - 1/\log_{10} S, 0.7\}$, which equals 0.67 at this application’s $S = 1,000$. Under that stricter form the largest United States value ($\hat{k} = 0.674$; Table D.1) would also cross the threshold, and the all-PV gate would withhold the United States estimates as well. The pre-committed rule fixed the 0.7 ceiling in advance, as does Jedhoff et al. (2025); the sensitivity of the gate decision to this choice is consistent with the variant’s role as a gated cross-check rather than a co-primary estimator.

The structural risk is visible in Equation (D.2): the stacked posterior is a geometric average over the M PV likelihoods, and each reweighting uses it as the proposal for a single component. When the per-PV posteriors are well separated relative to their spread (a separation that tends to increase as n grows), the ratios acquire heavy tails and $\hat{k}$ rises. A tempering exponent is available as a modification (replace $1/M$ by $1/(kM)$, $k = 2, 3, \dots$, flattening the proposal), and practical guidance is to run the untempered $1/M$ bridge first and consider tempering only if the gate check fails; we record this as guidance rather than as an evaluated remedy, since no tempered variant is part of the frozen evidence in this article.

The provenance of the reported estimates is as follows. The final reweighted-stack variant table was produced by the estimates script consuming the calibrated stack’s production CmdStan draws, and the evidence-packet manifest records `production_cdirect_proposal_used = TRUE` and `old_lme4_laplace_surrogate_promoted = FALSE`. A legacy lme4–Laplace surrogate proposal path remains in the repository as a development-era diagnostic and was never promoted to production evidence.

D.2. PISA 2022 Reliability Diagnostics. Figure 4 of the main text and Table D.1 report the per-PV Pareto $\hat{k}$ for both countries, computed from the production stacked draws with identical construction. The two countries fall on opposite sides of the gate: all ten United States PVs pass ($\hat{k}$ 0.347–0.674), and all ten Korea PVs fail ($\hat{k}$ 0.819–1.341), with PSIS effective sample sizes in Korea of roughly 5–30 draws out of 1,000, far too few to support reweighted inference.

TABLE D.1. Per-PV PSIS diagnostics for the reweighted-stack variant in the PISA 2022 application. n_{eff} is the PSIS effective sample size out of $S = 1,000$ proposal draws.

Country	PV	$\hat{k}$	n_{eff}	Status
Korea	1	0.999	10.0	unreliable
Korea	2	0.979	10.5	unreliable
Korea	3	0.835	20.3	unreliable
Korea	4	1.341	5.0	unreliable
Korea	5	1.217	12.0	unreliable
Korea	6	0.885	20.9	unreliable
Korea	7	0.819	29.7	unreliable
Korea	8	1.052	28.9	unreliable
Korea	9	0.965	9.8	unreliable
Korea	10	0.966	10.6	unreliable
United States	1	0.493	299.9	usable
United States	2	0.671	207.8	usable
United States	3	0.463	365.6	usable
United States	4	0.347	247.5	usable
United States	5	0.442	183.5	usable
United States	6	0.387	533.6	usable
United States	7	0.442	359.5	usable
United States	8	0.494	91.1	usable
United States	9	0.420	228.0	usable
United States	10	0.674	132.5	usable

D.3. United States: Reportable Estimates. With the gate passed, the United States estimates from the reweighted-stack variant are reportable; Table 5 of the main text reports the estimates. They lie close to the per-PV and calibrated-stack values of the main text even though the reweighting aims at the staged target of [Equation \(D.1\)](#) rather than the level-1 placement of the per-PV and calibrated-stack fits: the within-school effect differs from the per-PV workflow by -0.056 (standard-error ratio 0.996) and the between-school effect by -0.989 (standard-error ratio 1.129). We do not interpret these differences beyond noting that they are small relative to the reported standard errors and that they compare estimates aimed at different declared weighting targets; the reweighted-stack variant remains a cross-check, not a co-primary estimator.

D.4. Korea: Withheld by the Gate. All ten Korea reweightings fail the gate, so the Korean reweighted-stack estimate and standard-error cells are intentionally withheld everywhere in this article, and the reported status is *blocked by PSIS*. The

withholding does not mean the Korean data are defective, and it does not mean the per-PV workflow or the calibrated stack is affected; both are PSIS-free and fully reportable in Korea. It means the importance-reweighting route from this stacked proposal to the declared per-PV staged targets is unreliable in this application, which is the contingency the gate exists to catch.

D.5. Relation to the Simulation Study. The simulation study of the main text does not evaluate the reweighted-stack variant; the archived simulation tables contain only non-evidence placeholder rows for it (Section E.2). Its evidence in this article is therefore the PISA application, where the same construction was run in both countries and the gate decided differently (Section D.2).

Appendix E. Simulation Study: Full Documentation

This appendix documents the simulation study of Section 5 of the main text: the complete design grid, the estimators, the provenance and validation of the evidence base, per-condition results for both fixed-effect gradients, and the Monte Carlo error methodology.

E.1. Design Grid. The data-generating model is the two-level random-intercept structure of Equation (2) of the main text with generating fixed effects $\beta^W = 0.40$ and $\beta^B = 0.60$. Schools are sampled with probability proportional to size and students within schools, producing informative weights; plausible values are generated from the latent outcome under a Gaussian measurement model with reliability ρ_{PV} . Two design profiles mirror the reading intraclass correlations of the application countries (United States-like: outcome ICC 0.20, covariate ICC 0.15; Korea-like: 0.45 and 0.30). Table E.1 lists all 24 conditions.

TABLE E.1. Complete simulation design grid: 24 conditions. J = schools; $\bar{n}$ = mean school size; ρ_{PV} = PV reliability; M = PV count. Replications per condition follow the stratum: $R = 100$ (mainstream), 400 (small-school stress), 200 (PV sensitivity).

Condition	Stratum	Profile	ICC _y	J	$\bar{n}$	ICC _x	ρ_{PV}	M
T1-c6	Mainstream	KOR	0.45	200	20	0.30	0.90	10
T2-c1	Mainstream	USA	0.20	100	20	0.15	0.90	10
T2-c2	Mainstream	USA	0.20	50	10	0.15	0.90	10
T2-c3	Mainstream	USA	0.20	50	30	0.15	0.90	10
T2-c4	Mainstream	USA	0.20	100	10	0.15	0.90	10
T2-c5	Mainstream	USA	0.20	100	30	0.15	0.90	10
T2-c6	Mainstream	USA	0.20	200	10	0.15	0.90	10
T2-c7	Mainstream	USA	0.20	200	30	0.15	0.90	10
T3-c1	Mainstream	USA	0.20	200	20	0.15	0.90	10
T3-c3	Mainstream	USA	0.20	100	20	0.30	0.90	10
T3-c4	Mainstream	KOR	0.45	100	20	0.15	0.90	10
T5-c1	Mainstream	KOR	0.45	100	20	0.30	0.90	10
T1-c1	Small-school stress	USA	0.20	50	20	0.15	0.90	10
T4-c1	Small-school stress	KOR	0.45	50	20	0.30	0.90	10
Q7-c2	PV sensitivity	KOR	0.45	100	20	0.30	0.90	10
Q8-c1	PV sensitivity	KOR	0.45	100	10	0.30	0.90	10
SK-c1	PV sensitivity	USA	0.20	100	20	0.15	0.85	10
SM-c1	PV sensitivity	USA	0.20	100	20	0.15	0.90	5
SM-c2	PV sensitivity	USA	0.20	100	20	0.15	0.90	20
T4-c3	PV sensitivity	USA	0.20	100	20	0.15	0.85	5
T4-c4	PV sensitivity	USA	0.20	100	20	0.15	0.85	20
T4-c5	PV sensitivity	USA	0.20	100	20	0.15	0.95	5
T4-c6	PV sensitivity	USA	0.20	100	20	0.15	0.95	10
T4-c7	PV sensitivity	USA	0.20	100	20	0.15	0.95	20

E.2. Estimators, Provenance, and Validation. Each replication is fit by four estimators on identical data: the *oracle* benchmark (the generating values, used as the coverage reference), the *per-PV REML* assembly (restricted-maximum-likelihood fits pooled by the combining rules), the *per-PV MCMC* workflow, and the *calibrated stack*. Both fixed-effect coordinates, the within-school slope and the between-school slope, are recorded from every fit. The evidence base for the exhibits comprises 16,000 fit records: 4,000 replications across the 24 conditions, each fit by all four estimators at the full replication count of its condition.

The design was executed twice. A first production run (June 2026, on a single pinned x86_64 node) recorded the within-school slope as its focal estimand and carries the archive’s original convergence audit (every potential-scale-reduction statistic at

or below 1.040, no divergent transitions); its tables are preserved unchanged in the reproducibility archive ([Section G.2](#)). Because that archive stores one estimand per fit, the design was re-executed in July 2026 to record both fixed-effect coordinates from every fit, with the same frozen generative sources (the data-generating, sampling-weight, and seed-schedule files match the hashes of the frozen design record), the same seed schedule (root seed 20260514), and the same sampler configuration (four chains of 2,000 iterations with 1,000 warmup, adapt-delta 0.95, maximum tree depth 12), on a single arm64 machine with CmdStan 2.38.0. In the second run every estimator is complete at the full replication count of every condition, whereas the first run carried reduced per-PV MCMC footprints in six conditions. The exhibits in this article report the second run.

Three checks bind the two runs and the reported numbers. First, the second run reproduces the first run’s within-school results: across the 24 conditions and the three non-oracle estimators, the condition-level mean estimates differ by at most 0.0086, and the largest differences arise in $R = 100$ conditions, where the Monte Carlo standard error of a single condition mean is itself about 0.003. Second, convergence diagnostics were re-extracted for two replications of every condition by re-running those fits under the recorded seed schedule; the chains are deterministic given the seeds, the data, and the toolchain, so the re-extracted chains are the production chains, which is confirmed by the re-extracted point estimates matching the stored values exactly. Across the 558 re-extracted MCMC fits (the per-PV fits and the stacked fit of two replications in every condition), every potential-scale-reduction statistic is at or below 1.020, the minimum bulk effective sample size is 312, and no divergent transitions occurred. Third, the Bayesian per-PV estimates agree with the deterministic REML estimates, which involve no sampling, replication by replication across all 4,000 replications: the largest absolute difference is 0.0007 for the within-school slope and 0.011 for the between-school slope, small fractions of the corresponding standard errors.

The archived tables of the first run also contain placeholder rows for estimator variants that were planned during development but not implemented in production; these carry a non-evidence flag, are excluded from every count and summary in this article, and are documented in the archive inventory ([Section G.2](#)). The reweighted-stack variant is among them: the simulation does not evaluate it, and its evidence in this article is the PISA application (main text, Section 6.4).

E.3. Per-Condition Results. Table E.2 lists the per-condition results underlying Figure 2 and Table 3 of the main text: bias and empirical coverage of nominal 95% intervals, with Wilson 95% intervals (Wilson, 1927), for both fixed-effect gradients.

TABLE E.2. Per-condition simulation results for the within-school slope β^W (generating value 0.40) and the between-school slope β^B (0.60): bias and empirical coverage of nominal 95% intervals with Wilson 95% intervals.

Condition	Estimator	β^W		β^B	
		Bias	Coverage [Wilson 95%]	Bias	Coverage [Wilson 95%]
T1-c6	Per-PV (MCMC)	-0.0000	0.980 [0.930, 0.994]	-0.0029	0.990 [0.946, 0.998]
T1-c6	Per-PV (REML)	-0.0000	0.980 [0.930, 0.994]	-0.0029	0.970 [0.915, 0.990]
T1-c6	Calibrated stack	-0.0000	0.960 [0.902, 0.984]	-0.0040	0.970 [0.915, 0.990]
T2-c1	Per-PV (MCMC)	-0.0002	0.980 [0.930, 0.994]	-0.0168	0.920 [0.850, 0.959]
T2-c1	Per-PV (REML)	-0.0002	0.980 [0.930, 0.994]	-0.0168	0.920 [0.850, 0.959]
T2-c1	Calibrated stack	-0.0002	0.980 [0.930, 0.994]	-0.0164	0.920 [0.850, 0.959]
T2-c2	Per-PV (MCMC)	0.0110	0.970 [0.915, 0.990]	0.0232	0.970 [0.915, 0.990]
T2-c2	Per-PV (REML)	0.0110	0.970 [0.915, 0.990]	0.0232	0.970 [0.915, 0.990]
T2-c2	Calibrated stack	0.0110	0.960 [0.902, 0.984]	0.0237	0.960 [0.902, 0.984]
T2-c3	Per-PV (MCMC)	0.0001	0.960 [0.902, 0.984]	-0.0075	0.960 [0.902, 0.984]
T2-c3	Per-PV (REML)	0.0001	0.960 [0.902, 0.984]	-0.0076	0.960 [0.902, 0.984]
T2-c3	Calibrated stack	0.0002	0.940 [0.875, 0.972]	-0.0082	0.960 [0.902, 0.984]
T2-c4	Per-PV (MCMC)	-0.0000	0.970 [0.915, 0.990]	-0.0017	0.930 [0.863, 0.966]
T2-c4	Per-PV (REML)	-0.0000	0.970 [0.915, 0.990]	-0.0017	0.920 [0.850, 0.959]
T2-c4	Calibrated stack	0.0000	0.970 [0.915, 0.990]	-0.0011	0.920 [0.850, 0.959]
T2-c5	Per-PV (MCMC)	0.0019	0.970 [0.915, 0.990]	-0.0182	0.970 [0.915, 0.990]
T2-c5	Per-PV (REML)	0.0019	0.970 [0.915, 0.990]	-0.0181	0.970 [0.915, 0.990]
T2-c5	Calibrated stack	0.0020	0.940 [0.875, 0.972]	-0.0174	0.960 [0.902, 0.984]
T2-c6	Per-PV (MCMC)	0.0000	0.990 [0.946, 0.998]	0.0125	0.960 [0.902, 0.984]
T2-c6	Per-PV (REML)	0.0000	0.990 [0.946, 0.998]	0.0126	0.960 [0.902, 0.984]
T2-c6	Calibrated stack	-0.0000	0.980 [0.930, 0.994]	0.0126	0.920 [0.850, 0.959]
T2-c7	Per-PV (MCMC)	0.0018	1.000 [0.963, 1.000]	-0.0004	0.970 [0.915, 0.990]
T2-c7	Per-PV (REML)	0.0018	1.000 [0.963, 1.000]	-0.0004	0.960 [0.902, 0.984]
T2-c7	Calibrated stack	0.0018	0.990 [0.946, 0.998]	-0.0001	0.970 [0.915, 0.990]
T3-c1	Per-PV (MCMC)	-0.0011	0.980 [0.930, 0.994]	0.0070	0.960 [0.902, 0.984]
T3-c1	Per-PV (REML)	-0.0011	0.980 [0.930, 0.994]	0.0070	0.960 [0.902, 0.984]
T3-c1	Calibrated stack	-0.0011	0.940 [0.875, 0.972]	0.0069	0.950 [0.888, 0.978]
T3-c3	Per-PV (MCMC)	0.0032	0.950 [0.888, 0.978]	-0.0137	0.960 [0.902, 0.984]
T3-c3	Per-PV (REML)	0.0032	0.950 [0.888, 0.978]	-0.0135	0.960 [0.902, 0.984]

Condition	Estimator	β^W		β^B	
		Bias	Coverage [Wilson 95%]	Bias	Coverage [Wilson 95%]
T3-c3	Calibrated stack	0.0032	0.940 [0.875, 0.972]	-0.0128	0.960 [0.902, 0.984]
T3-c4	Per-PV (MCMC)	0.0011	0.960 [0.902, 0.984]	-0.0050	0.980 [0.930, 0.994]
T3-c4	Per-PV (REML)	0.0011	0.960 [0.902, 0.984]	-0.0051	0.980 [0.930, 0.994]
T3-c4	Calibrated stack	0.0012	0.950 [0.888, 0.978]	-0.0064	0.980 [0.930, 0.994]
T5-c1	Per-PV (MCMC)	-0.0021	0.970 [0.915, 0.990]	-0.0144	0.910 [0.838, 0.952]
T5-c1	Per-PV (REML)	-0.0021	0.970 [0.915, 0.990]	-0.0143	0.910 [0.838, 0.952]
T5-c1	Calibrated stack	-0.0020	0.960 [0.902, 0.984]	-0.0140	0.910 [0.838, 0.952]
T1-c1	Per-PV (MCMC)	0.0000	0.985 [0.968, 0.993]	0.0034	0.958 [0.933, 0.973]
T1-c1	Per-PV (REML)	0.0000	0.985 [0.968, 0.993]	0.0036	0.953 [0.927, 0.969]
T1-c1	Calibrated stack	0.0001	0.973 [0.951, 0.985]	0.0044	0.955 [0.930, 0.971]
T4-c1	Per-PV (MCMC)	0.0010	0.988 [0.971, 0.995]	0.0054	0.945 [0.918, 0.963]
T4-c1	Per-PV (REML)	0.0010	0.988 [0.971, 0.995]	0.0055	0.938 [0.909, 0.957]
T4-c1	Calibrated stack	0.0010	0.980 [0.961, 0.990]	0.0058	0.950 [0.924, 0.967]
Q7-c2	Per-PV (MCMC)	0.0001	0.985 [0.957, 0.995]	-0.0037	0.965 [0.930, 0.983]
Q7-c2	Per-PV (REML)	0.0001	0.985 [0.957, 0.995]	-0.0034	0.960 [0.923, 0.980]
Q7-c2	Calibrated stack	0.0001	0.975 [0.943, 0.989]	-0.0044	0.960 [0.923, 0.980]
Q8-c1	Per-PV (MCMC)	-0.0044	0.980 [0.950, 0.992]	-0.0039	0.960 [0.923, 0.980]
Q8-c1	Per-PV (REML)	-0.0045	0.980 [0.950, 0.992]	-0.0041	0.955 [0.917, 0.976]
Q8-c1	Calibrated stack	-0.0045	0.960 [0.923, 0.980]	-0.0045	0.955 [0.917, 0.976]
SK-c1	Per-PV (MCMC)	0.0003	0.985 [0.957, 0.995]	0.0093	0.960 [0.923, 0.980]
SK-c1	Per-PV (REML)	0.0003	0.985 [0.957, 0.995]	0.0093	0.960 [0.923, 0.980]
SK-c1	Calibrated stack	0.0004	0.980 [0.950, 0.992]	0.0094	0.955 [0.917, 0.976]
SM-c1	Per-PV (MCMC)	0.0011	0.965 [0.930, 0.983]	-0.0101	0.950 [0.910, 0.973]
SM-c1	Per-PV (REML)	0.0011	0.965 [0.930, 0.983]	-0.0101	0.945 [0.904, 0.969]
SM-c1	Calibrated stack	0.0011	0.950 [0.910, 0.973]	-0.0102	0.950 [0.910, 0.973]
SM-c2	Per-PV (MCMC)	-0.0015	0.985 [0.957, 0.995]	-0.0031	0.950 [0.910, 0.973]
SM-c2	Per-PV (REML)	-0.0015	0.985 [0.957, 0.995]	-0.0031	0.945 [0.904, 0.969]
SM-c2	Calibrated stack	-0.0015	0.970 [0.936, 0.986]	-0.0034	0.945 [0.904, 0.969]
T4-c3	Per-PV (MCMC)	0.0002	0.990 [0.964, 0.997]	-0.0001	0.975 [0.943, 0.989]
T4-c3	Per-PV (REML)	0.0002	0.990 [0.964, 0.997]	-0.0001	0.970 [0.936, 0.986]
T4-c3	Calibrated stack	0.0002	0.975 [0.943, 0.989]	0.0003	0.965 [0.930, 0.983]
T4-c4	Per-PV (MCMC)	0.0010	0.990 [0.964, 0.997]	0.0133	0.950 [0.910, 0.973]
T4-c4	Per-PV (REML)	0.0010	0.990 [0.964, 0.997]	0.0133	0.950 [0.910, 0.973]
T4-c4	Calibrated stack	0.0010	0.980 [0.950, 0.992]	0.0133	0.945 [0.904, 0.969]
T4-c5	Per-PV (MCMC)	-0.0030	0.945 [0.904, 0.969]	-0.0002	0.955 [0.917, 0.976]
T4-c5	Per-PV (REML)	-0.0030	0.945 [0.904, 0.969]	-0.0001	0.945 [0.904, 0.969]
T4-c5	Calibrated stack	-0.0030	0.930 [0.886, 0.958]	0.0003	0.955 [0.917, 0.976]
T4-c6	Per-PV (MCMC)	0.0002	0.965 [0.930, 0.983]	-0.0071	0.965 [0.930, 0.983]

Condition	Estimator	β^W		β^B	
		Bias	Coverage [Wilson 95%]	Bias	Coverage [Wilson 95%]
T4-c6	Per-PV (REML)	0.0002	0.960 [0.923, 0.980]	-0.0070	0.950 [0.910, 0.973]
T4-c6	Calibrated stack	0.0002	0.955 [0.917, 0.976]	-0.0073	0.955 [0.917, 0.976]
T4-c7	Per-PV (MCMC)	0.0009	0.945 [0.904, 0.969]	-0.0173	0.960 [0.923, 0.980]
T4-c7	Per-PV (REML)	0.0009	0.945 [0.904, 0.969]	-0.0172	0.960 [0.923, 0.980]
T4-c7	Calibrated stack	0.0009	0.940 [0.898, 0.965]	-0.0175	0.965 [0.930, 0.983]

E.4. Monte Carlo Error Methodology. Stratum-level bias values carry Monte Carlo standard errors computed as the standard deviation of the per-fit estimates divided by the square root of the number of fits. Coverage uncertainty is reported through Wilson 95% intervals, which reflect each condition’s replication count; conditions with $R = 400$ resolve coverage to roughly ± 0.02 and conditions with $R = 100$ to roughly ± 0.04 at the observed rates. Readers comparing estimators within a condition should note that all estimators share the same replication data, so between-estimator differences are paired rather than independent.

Appendix F. PISA 2022 Application: Data Construction and Secondary Analyses

This appendix documents the empirical application of Section 6 of the main text: the analytic-sample cascade, the exact weight construction, sampler settings and diagnostics, and a qualified secondary analysis of the between-country contrast’s sensitivity to weight placement.

F.1. Data and Analytic-Sample Cascade. The analysis uses the public PISA 2022 student files (OECD, 2024b) for the United States and Korea, reading domain. The outcome variables are the ten reading plausible values PV1READ–PV10READ (complete for all sampled students); the covariate is the ESCS index; the cluster identifier is the PISA school ID; and the weight variables are the final student weight W_FSTUWT, the school base weight W_SCHGRNRABWT, and the 80 Fay-adjusted replicate weights (OECD, 2024a) (used only in the design-based target variant of Section C.3, not in the main-text analysis).

TABLE F.1. Analytic-sample cascade. PVs and replicate weights are complete for all released students; the only exclusion is missing ESCS.

	United States		Korea	
	Students	Schools	Students	Schools
Released country sample	4,552	154	6,454	186
Missing ESCS (excluded)	210	—	63	—
Analytic sample	4,342	150	6,391	186

Removing ESCS-missing students leaves four United States schools with no remaining sampled students, which is why the school count falls from 154 to 150; no Korean school is emptied. The ESCS decomposition uses weighted school means, $\bar{x}_{\cdot j} = \sum_{i \in j} w_{ij} x_{ij} / \sum_{i \in j} w_{ij}$, computed on the analytic sample with the normalized weights defined next; $x_{ij} - \bar{x}_{\cdot j}$ enters as the within-school regressor and $\bar{x}_{\cdot j}$ as the between-school regressor.

F.2. Weight Construction. The per-PV and calibrated-stack workflows weight student i 's log-likelihood contribution by a final student weight normalized to sum to the sample size within country, following the pseudo-posterior convention (Savitsky & Toth, 2016; Williams & Savitsky, 2020); the reweighted-stack variant's diagnostic reweighting targets a different object, the declared SNS-01 staged marginal pseudo-likelihood, described in Section D.1. The weight construction retains the declared two-stage decomposition, which supplies the reweighted-stack variant's staged target as well as the diagnostics and the secondary analysis below. `W_FSTUWT` is the codebook-designated full-sample weight for student-level analyses (OECD, 2024a), and `W_SCHGRNRABWT` is the school base weight released in the school file. With $S_j = \text{W_SCHGRNRABWT}$ (constant within school) and $I_{ij} = \text{W_FSTUWT} / \text{W_SCHGRNRABWT}$ (the conditional student weight), define

$$\bar{S} = \frac{1}{J} \sum_{j=1}^J S_j, \quad c_A = \frac{n}{\sum_{i,j} S_j I_{ij}}, \quad w_j^{\text{sch}} = \frac{S_j}{\bar{S}}, \quad w_{ij}^{\text{cond}} = c_A \bar{S} I_{ij}, \quad (\text{F.1})$$

so that the analysis weight is

$$w_{ij} = w_j^{\text{sch}} w_{ij}^{\text{cond}} = c_A S_j I_{ij} = \frac{n \cdot \text{W_FSTUWT}_{ij}}{\sum_{i,j} \text{W_FSTUWT}_{ij}},$$

the final student weight scaled to sum to the analytic sample size (equivalently, mean one). The school factor w_j^{sch} averages to one across schools and the conditional factor absorbs the remaining scale; the reconstruction $w_j^{\text{sch}} w_{ij}^{\text{cond}} \propto \text{W_FSTUWT}_{ij}$ is verified

to numerical precision in the data-build checks. Replicate weights and the raw final weight are firewalled out of the reweighted-stack variant’s importance-ratio numerators by contract checks that fail hard if violated. This level-1 placement of the (normalized) final weight is a declared analytic choice for the per-PV and calibrated-stack workflows, so that their comparison is never confounded with weighting choices. The staged placement used by the reweighted-stack variant’s targets, in which the school factor multiplies each school’s integrated likelihood contribution and the conditional student factor enters within that integral (the staged multilevel pseudolikelihood form of Rabe-Hesketh and Skrondal 2006), is a deliberate feature of the diagnostic design documented in Section D.1. The sensitivity of a secondary between-country contrast to such placement choices is examined in Section F.4.

F.3. Sampler Settings and Diagnostics. The per-PV workflow ran 20 per-PV fits ($2 \text{ countries} \times 10 \text{ PVs}$) of the weighted random-intercept model in Stan (Carpenter et al., 2017), each with two chains of 500 warmup iterations and 500 post-warmup draws, for 1,000 posterior draws from each per-PV fit, with adapt-delta 0.95, maximum tree depth 12, and a deterministic seed schedule keyed by country, workflow, and PV. The calibrated stack ran one stacked fractional fit per country under the same settings, retaining $S = 1,000$ draws for calibration (and, downstream, as the reweighted-stack variant’s proposal). All 22 fits completed with zero divergent transitions. Across the per-PV workflow fits the maximum $\hat{R}$ was 1.016 with minimum bulk effective sample size ≈ 198 ; the calibrated stack fits reached maximum $\hat{R}$ values of 1.006 (United States) and 1.008 (Korea) with minimum bulk effective sample sizes of 243 and 404. The per-PV workflow interval estimates use the student- t reference with Barnard–Rubin degrees of freedom (Barnard & Rubin, 1999), with per-estimand values 350.0 and 1360.2 for the United States within- and between-school effects, and 64.6 and 536.8 for Korea; the calibrated stack intervals are normal-reference intervals formed from the calibrated mean and covariance, which is why interval endpoints in Table 4 of the main text differ from the per-PV workflow’s by less than 0.1 score point (a t versus a normal reference) while the estimates and standard errors agree exactly.

F.4. Secondary Analysis: Weight-Placement Sensitivity of the Between-Country Contrast. The main text deliberately draws no conclusion about the United States–Korea difference in the between-school ESCS effect. This subsection documents why, as a qualified secondary analysis. The derived contrast $\Delta =$

$\beta_{\text{KOR}}^B - \beta_{\text{USA}}^B$ is not a target estimand of the study; it is scaling-sensitive, and the frozen record marks it *qualified, scaling-sensitive; strong claims disallowed*.

Two weighting schemes for the same model and data give materially different impressions of Δ (Table F.2, Figure F.1). Under the level-1 placement of the normalized final student weight (the declared scheme of the primary analysis), the contrast is large and nominally significant. Under a declared two-stage placement that assigns the school base weight to the school level and the conditional student weight to the student level, the contrast attenuates by more than half and is no longer significant at conventional levels. Both rows use deterministic per-PV fits with replicate-weight standard errors pooled by the combining rules; because the public files release replicate weights for the final student weight only, the two-stage row’s replicates perturb the conditional student factor while holding the school factor fixed, a declared allocation recorded with the archived results (the official stage-specific replicate decomposition is not identified from the public files, Section C.3). The two schemes answer subtly different population questions, and nothing in this article adjudicates between them; the point of record is that a cross-country comparison of between-school gradients can depend on weight placement, so we decline to make one. A robustness grid over weighting variants, weight scalings, PV subsets, and degrees-of-freedom conventions, archived with the frozen results, shows the contrast’s nominal significance depending on these choices, which is the basis for the frozen record’s qualification. (For completeness: the primary per-PV/calibrated-stack analyses imply a descriptive level-1-conditional gap of $92.94 - 69.74 = 23.20$, consistent with the level-1 row of Table F.2; it inherits the same qualification.)

TABLE F.2. Weight-placement sensitivity of the derived between-country contrast $\Delta = \beta_{\text{KOR}}^B - \beta_{\text{USA}}^B$ (PISA 2022 reading, deterministic design-based per-PV fits pooled by the combining rules). Secondary, qualified analysis; no strong claim is licensed.

Weighting scheme	$\hat{\Delta}$	SE	df	p
Final student weight only (level-1 placement)	23.58	4.28	73.2	$p < .001$
Declared two-stage weights (school and conditional student)	10.66	6.53	113.0	$p = .105$

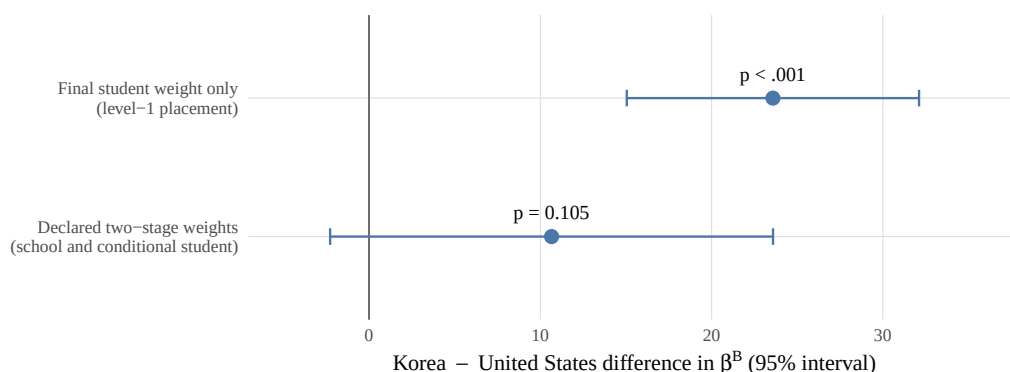


FIGURE F.1. The derived Korea–United States difference in the between-school ESCS effect under two weight-placement schemes, with 95% intervals. The contrast is scaling-sensitive: nominally significant under level-1-only placement of the final student weight, attenuated and nonsignificant under the declared two-stage placement.

F.5. Contract and Negative Controls. The empirical analysis ran under a frozen estimand contract (`PISA2022_READ_EST01_PHASE7R_ABC_WITHIN_BETWEEN`) that fixes the model formula, the PV set and order, the weighting scheme, the co-primary estimands (β^W , β^B per country), the rule that country-gap contrasts are secondary-only, and the per-workflow reporting gates (including the reweighted-stack variant’s PSIS gate and the withhold rule). Contract negative controls verify that violations fail hard: dropping either estimand, promoting the country gap to primary, or feeding final or replicate weights into the reweighted-stack variant’s importance-ratio numerator each raises an error rather than producing output. Row-support and weight-metadata hashes bind the analytic sample and weight construction to the reported results ([Section G](#)).

Appendix G. Software and Reproducibility Inventory

G.1. The R Package. The workflow is implemented in the open-source R package `pvstackr` (version 0.2.0; the repository reference appears on the title page). The package exposes a declared-design interface in which the analyst’s choices are recorded as objects before any fitting occurs:

- `pv_design()` declares the PV columns, replicate-weight columns, and Fay factor of the survey design;

- `pv_brr_target()` assembles the external fixed-effect target of [Section C](#) from deterministic per-PV (and, optionally, per-replicate) refits, recording provenance hashes;
- `pv_fit()` runs one of the three methods, `"per_pv"` (the per-PV workflow), `"stack_psis"` (the reweighted-stack variant), or `"stack_direct"` (the calibrated stack), against a declared design and target;
- `pv_compare_methods()` aligns the outputs of several methods on matched estimands with agreement diagnostics;
- `accessors` (`get_estimates()`, `get_target()`, `get_diagnostics()`, `get_draws()`) return the estimates table (with interval metadata and fraction of missing information), the target object and its provenance, sampler and calibration diagnostics, and the calibrated draws.

Per-row reporting metadata records whether each interval is coverage-claimable under the article’s evidence, and the reweighted-stack variant methods implement the gate-and-withhold rule of [Section D](#) as package behavior, not as user discretion. Ten vignettes (five applied, five methodological) document the workflow end to end. Analyses in the article used R (R Core Team, [2026](#)) with Stan via `cmdstanr` (Carpenter et al., [2017](#); Gabry et al., [2024](#)), `lme4` (Bates et al., [2015](#)) for deterministic refits, and `loo` (Vehtari et al., [2025](#)) for PSIS.

G.2. Evidence Archive. Every number reported in the article resolves to an archived evidence table.

- *Simulation evidence tables.* Per-fit, per-condition, and stratum-level tables for the 24-condition study, generated under a single root seed (20260514) and a deterministic seed schedule keyed by condition, replication, and estimator, with the fixed MCMC configuration of [Section E.2](#). Two table sets are archived: the June 2026 run (within-school estimand; original convergence audit) and the July 2026 both-gradient run that the exhibits report, together with the reproduction and diagnostics checks of [Section E.2](#). The June tables also contain non-evidence placeholder rows for estimator variants planned during development but not implemented in production; these are flagged and excluded from every summary.
- *Empirical evidence tables.* The PISA estimates and reliability diagnostics are emitted as a versioned packet in which every row carries a hash of its source artifact; a verifier script and an automated test suite regenerate and re-check the packet.

- *Label mapping.* The archived tables predate the workflow names used in this article. The correspondence is as follows: `bayes_a` = per-PV (MCMC); `freq_per_pv` = per-PV (REML); `c_direct` = calibrated stack; `pipeline_b` = reweighted-stack variant; `oracle` = oracle benchmark.

G.3. Data Availability. PISA 2022 data are publicly available from the OECD; the archive contains the data-build scripts (country extraction, ESCS decomposition, weight construction with the reconstruction checks of [Section F.2](#), and the analytic-sample cascade of [Table F.1](#)) but not the data files themselves. The simulation is fully self-contained and regenerates its data from seeds.

G.4. Replication Package. The analysis engine, the evidence archive, an automated verifier and test suite, and a rendered reproduction guide are distributed together as a public replication package (repository on the title page). Its default path regenerates every table and figure in the article and this supplement from the included evidence in minutes, with no model refitting; a verification track re-checks the artifact manifest and hashes, the registered numeric anchors, the reporting gates, and the publication-hygiene scans, and fails closed. Opt-in tracks expose the intermediate calculations, a reduced simulation replay, the full 4,000-replication simulation protocol, and the 22-fit PISA protocol. Unit-record PISA files are not redistributed; the package supplies the data-build scripts and an authorized-access contract instead.